FOREIGN TRADE UNIVERSITY

HO CHI MINH CITY CAMPUS

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GRADUATION THESIS

Major: International Finance

BITCOIN VOLATILITY AND HEDGING EFFECTIVENESS UNDER GLOBAL UNCERTAINTY: A STRUCTURAL-BREAK DCC-MIDAS-X APPROACH

THESIS ID: B41

Student name: Nguyễn Hải Duy

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STATEMENT OF AUTHORSHIP

I hereby declare that this graduation thesis, entitled “Bitcoin Volatility and Hedging Effectiveness under Global Uncertainty: A Structural-Break DCC-MIDAS-X Approach,” is my own original work. Except where proper references are provided, all ideas, analyses, and findings presented in this thesis are the result of my independent research.

I further confirm that this thesis has not been submitted, either in whole or in part, for any degree or academic qualification at any other institution. All sources of information and contributions from others have been fully acknowledged. I take full responsibility for the accuracy, integrity, and authenticity of the content presented in this thesis.

Ho Chi Minh City, June 2026

Student Author

Nguyen Hai Duy

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While conducting this research, I have strived to structure it as effectively as possible. However, I acknowledge that perfection is unattainable, and there may be areas for improvement. I warmly welcome any constructive feedback and suggestions from lecturers and readers to further refine this study.

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LIST OF ABBREVIATIONS

No.	Abbreviation	Description
1	ADF	Augmented Dickey–Fuller
2	AIC	Akaike Information Criterion
3	ARCH	Autoregressive Conditional Heteroskedasticity
4	BEKK	Baba–Engle–Kraft–Kroner
5	BIC	Bayesian Information Criterion
6	BTC	Bitcoin
7	CAPM	Capital Asset Pricing Model
8	CCC	Constant Conditional Correlation
9	CME	Chicago Mercantile Exchange
10	CPI	Consumer Price Index
11	CW	Clark–West
12	DCC	Dynamic Conditional Correlation
13	DCC-MIDAS	Dynamic Conditional Correlation–Mixed Data Sampling
14	DM	Diebold–Mariano
15	EMU	European Monetary Union
16	EPU	Economic Policy Uncertainty
17	ETF	Exchange-Traded Fund
18	GARCH	Generalized Autoregressive Conditional Heteroskedasticity
19	GARCH-MIDAS	GARCH–Mixed Data Sampling

20	GDP	Gross Domestic Product
21	GEPU	Global Economic Policy Uncertainty
22	HE	Hedging Effectiveness
23	IMF	International Monetary Fund
24	JB	Jarque–Bera
25	LR	Likelihood Ratio
26	MIDAS	Mixed Data Sampling
27	MSCI	Morgan Stanley Capital International
28	MSPE	Mean Squared Prediction Error
29	NLS	Non-Linear Least Squares
30	OHR	Optimal Hedge Ratio
31	OLS	Ordinary Least Squares
32	PIMCO	Pacific Investment Management Company
33	PP	Phillips–Perron
34	PPP	Purchasing Power Parity
35	QML	Quasi-Maximum Likelihood
36	S&P GSCI	Standard & Poor's Goldman Sachs Commodity Index
37	SB	Structural Break
38	SEC	U.S. Securities and Exchange Commission
39	VIX	CBOE Volatility Index

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CHAPTER 1: INTRODUCTION

1.1 Rationale

Over the past decade, global financial markets have absorbed a relentless sequence of macroeconomic shocks, including Brexit (2016), the US–China trade war (2018–2019), the COVID-19 crisis (2020), the Russia–Ukraine war (2022), the 2022–2024 monetary tightening cycle, and renewed tariff and trade-policy tensions (IMF, 2025). These events inevitably pushed Global Economic Policy Uncertainty (GEPU) to extreme levels[1], systematically destabilizing broad asset markets (Su et al., 2025). More importantly, what made these episodes so damaging was not simply the size of the losses; it was the speed and persistence with which policy uncertainty reshaped cross-asset correlations, forcing investors to question whether traditional diversification rules could still provide reliable portfolio protection (Grothe et al., 2025).

More critically, the deeper problem lay in the failure of the very hedging instruments investors had long trusted for portfolio defense. In principle, high-grade fixed income acts as a shock absorber: when equities plummet, government bonds and investment-grade corporate debt hold their value or appreciate, limiting portfolio drawdowns (Baur & Lucey, 2010). Yet in the sharpest stress episodes of recent years, this stabilizing relationship inverted — investment-grade bonds and global equities fell in tandem (Akhtaruzzaman et al., 2021), inflicting simultaneous losses precisely when diversification was most needed (McQuinn, Hyde & O'Brien, 2025). Diversification's core promise of protection through imperfectly correlated assets (Harry Markowitz, 1952) thus held only while correlations remained stable, breaking down precisely in crises.

In response to this breakdown of traditional hedges, institutional asset managers began looking beyond conventional markets, increasingly treating Bitcoin as a potential portfolio instrument (Fidelity Digital Assets, 2024). What began as an

[1] GEPU is a GDP-weighted global index developed by Baker, Bloom, and Davis (2016) that captures the intensity of policy-related uncertainty across major economies, turning dispersed concerns over fiscal, monetary, trade, and regulatory decisions into a single measure of global policy risk.

obscure cryptographic experiment in 2009 (Nakamoto, 2008) had, by the early 2020s, grown into a trillion-dollar asset class (CoinMarketCap, 2025), cemented by the SEC's approval of spot Bitcoin ETFs in January 2024[2] and the entry of publicly listed corporations holding Bitcoin on their balance sheets (Aufiero et al., 2025). No longer a fringe bet for retail speculators, Bitcoin had become a deliberate allocation decision for professional money managers, making its capacity to reduce portfolio risk directly relevant to institutional risk management (Bouri et al., 2021).

However, the empirical evidence on Bitcoin's hedging capacity remains mixed and strongly sample-dependent. For instance, studies of calmer periods generally found Bitcoin largely uncorrelated with equities and bonds, pointing to real diversification benefits (Stensås et al., 2019; Platanakis & Urquhart, 2020), whereas crisis-window evidence is more cautious as its extreme volatility undermined hedging suitability even in normal conditions (Smales, 2019), and during the COVID-19 crash Bitcoin collapsed alongside global equities, showing limited safe-haven capacity at the time of greatest need for downside protection (Goodell & Goutte, 2021). These conflicting findings reflect the framework of Baur and Lucey (2010), whose three roles—hedge, safe haven, and diversifier—are conditional rather than fixed. Bitcoin's hedging capacity is therefore regime-dependent rather than a stable portfolio property, which explains why findings vary across samples (Xu & Kinkyo, 2023).

Fundamentally, this inconsistency stems not from Bitcoin's properties being unknowable, but from a flawed assumption embedded in the standard models. Most of the cryptocurrency hedging literature relies on the Dynamic Conditional Correlation (DCC) model (Engle, 2002), which allows correlations to evolve only gradually. While DCC captures short-term volatility dynamics well, the model lacks the capacity to capture structural shifts in co-movements driven by macroeconomic transitions (Fang et al., 2019; Conlon & McGee, 2020). During systemic shocks, such as the COVID-19 pandemic or the 2022 tightening cycle, a single-regime

[2] On January 10, 2024, the SEC approved eleven U.S.-listed spot Bitcoin ETFs, granting investors regulated exchange-traded access to Bitcoin without direct custody — marking its formal integration into the regulated financial system and accelerating the shift in its investor base from retail to institutional.

specification can misread persistent structural breaks as temporary shocks, producing biased hedge ratios giving investors false protection signals (Xiong et al., 2024).

To overcome these non-linear limitations, recent econometric advances introduce a dual-component volatility framework. For example, the GARCH-MIDAS framework (Engle et al., 2013) and its bivariate extension, DCC-MIDAS (Colacito et al., 2011), separate volatility and correlation into two layers: a short-run component that captures high-frequency daily return fluctuations, and a long-run component driven by low-frequency macroeconomic forces. By connecting this long-run component to macro variables like GEPU or VIX through a Mixed-Data Sampling weighting scheme, the model can capture and forecast how sustained global policy uncertainty shapes the persistent structure of asset co-movements (Hu & Borjigin, 2024). Empirically, Fang et al. (2019) were the first to apply such framework to Bitcoin, pairing GARCH-MIDAS for volatility with DCC-MIDAS for its correlations with equities and commodities, with GEPU as the macro driver. Importantly, their estimates show that GEPU significantly drives Bitcoin's long-run volatility and correlations, while GEPU-conditioned hedge ratios deliver modest but measurable improvements in hedging performance relative to standard DCC benchmarks.

Yet Fang et al. (2019) made a critical simplifying assumption: that Bitcoin's sensitivity to GEPU remained constant throughout their sample, which ended in early 2018. The profound macroeconomic upheavals of the post-2018 era fundamentally invalidate this static baseline. The trade war, the pandemic, the tightening cycle, and the ETF approval represent four qualitatively different environments for Bitcoin (Wątorek et al., 2023) — each with its own investor base, risk appetite, and transmission mechanism from macro uncertainty to asset prices (Akhtaruzzaman et al., 2021; Bouri et al., 2022). If GEPU's effect on Bitcoin's volatility and correlations shifted across these volatile regimes, then treating the relationship as a single fixed parameter would blur distinct regime-specific relationships (Ang & Timmermann, 2012; Bai & Perron, 2003). Consistent with this, Xiong et al. (2024) showed that adding Bai–Perron structural breaks to DCC-MIDAS substantially improves model fit and portfolio risk reduction in a stock–exchange-rate setting, yet their framework is applied outside the cryptocurrency context and does not decompose whether the gain arises from macroeconomic conditioning, regime accommodation, or their interaction.

In light of the foregoing, this thesis, "Bitcoin Volatility and Hedging Effectiveness under Global Uncertainty: A Structural-Break DCC-MIDAS-X Approach", aims to bridge the gap in the literature as well as contribute up-to-date empirical evidence on the regime-dependent transmission of global economic policy uncertainty into Bitcoin's long-run risk dynamics. The study develops a GARCH-MIDAS-X and DCC-MIDAS-X framework that incorporates four endogenously identified GEPU regimes, providing a comprehensive assessment of Bitcoin's hedging effectiveness across major asset classes. By adapting to regime-specific structural uncertainty rather than just daily price fluctuations, the resulting macro-sensitive hedge ratios offer actionable guidance for institutional investors seeking to protect multi-asset portfolios from recurring systemic shocks.

1.2 Research background

1.2.1 Bitcoin's role in financial markets

Launched in 2009 by Satoshi Nakamoto, Bitcoin emerged as a decentralized digital currency independent from sovereign governments, centralized institutions, and banking systems, and is often seen as a response to the limitations of traditional financial structures (Demir et al., 2018). In its early years, Bitcoin attracted limited institutional attention and was traded mainly by retail participants and speculators (JPMorgan Chase & Co., 2021). Over the following decade, however, Bitcoin's financial status changed substantially as cryptocurrency markets developed their own systematic risk factors and became increasingly integrated into asset-pricing research (Liu et al., 2022). This transition was reinforced by major milestones, including the launch of regulated Bitcoin futures on CBOE and CME in December 2017, the approval of Bitcoin futures ETFs in 2021, and the approval of spot Bitcoin ETFs in January 2024 (Menchetti et al., 2021; Gensler, 2024). At the same time, Bitcoin's market capitalization expanded dramatically, rising by more than 800% from about USD 216 billion at the end of 2017 to over USD 2 trillion by early 2026, while listed corporations such as MicroStrategy and Tesla began allocating cash reserves to Bitcoin (Aufiero et al., 2025). Bitcoin therefore moved beyond its original retail-speculative niche and became a deliberate portfolio allocation question for professional investors (Fidelity Digital Assets, 2024).

Even as large institutions have adopted it, the debate over Bitcoin's finance and economics continues to intensify. The main reasons are its speculative nature (Borisov, 2024; Sanz Bas, 2024), its repeated price bubbles (Gemici et al., 2023; Podhorsky, 2024), its very large price swings (S&P Global, 2026), and the growing concerns over regulation, market manipulation, cyberattacks, and weak exchange security (Fratrič et al., 2022; Náñez Alonso et al., 2024). A separate line of research keeps warning that Bitcoin can be misused — for money laundering through mixers and similar tools (Chainalysis, 2024; Rosenquist et al., 2024) or in Ponzi-style scams such as OneCoin and BitConnect (Agarwal et al., 2024; CoinLedger, 2025) — while other studies still argue that Bitcoin behaves mostly like a risky, speculative asset rather than a stable means of payment tied to the wider economy (Borisov, 2024). Despite all this controversy, Bitcoin has continued to attract institutional attention and remain actively traded across global financial markets, making its risk profile increasingly relevant to professional portfolio management (Liu et al., 2022). It is exactly this mix of rapid growth and sharp volatility that has made measuring and modelling Bitcoin's price volatility a key research topic, closely tied to investment and risk-management decisions (Mostafa et al., 2021).

As Bitcoin moved from a speculative digital asset into an institutionally traded portfolio instrument, the relevant question also became more demanding (Baur et al., 2018; Guesmi et al., 2019). It is no longer sufficient to ask whether Bitcoin has low average correlation with traditional assets. What matters is whether that low co-movement persists when market conditions change, since the distinction between a hedge, a safe haven, and a diversifier is meaningful only if the underlying correlation structure is stable enough to generate actual risk reduction (Urquhart & Zhang, 2019; Shahzad et al., 2020). This issue is particularly important for Bitcoin because institutional participation can strengthen its exposure to common risk factors, making its diversification value more conditional on the prevailing macro-financial regime (Umar et al., 2022). The empirical literature therefore shifted from treating Bitcoin as a structurally independent asset toward testing whether its hedging role varies across samples, asset classes, and periods of market stress (Kumar et al., 2025).

Consistently, a parallel literature on Bitcoin's diversification capacity produced a broadly optimistic view, supported by a wide range of empirical methods: vector autoregressive and distributed-lag models (Panagiotidis et al., 2018), wavelet coherence (Kraaijeveld & De Smedt, 2020), time-frequency connectedness (Milunovich, 2018), directed acyclic graphs (Kumar & Anandarao, 2019), and univariate and multivariate GARCH models (Stensås et al., 2019) all found Bitcoin only weakly correlated with equities, bonds, and commodities, implying real diversification benefits (Platanakis & Urquhart, 2020). The prevailing rationale was that Bitcoin behaves independently of conventional financial developments because it is driven by a distinct set of non-financial forces — social sentiment (Kristoufek, 2015), mining-related energy prices (Walther et al., 2019), user anonymity (Foley et al., 2019), and programming-community engagement (Kraaijeveld & De Smedt, 2020). Yet this independence proved conditional rather than structural. Panagiotidis et al. (2019) showed that macro-financial fundamentals significantly affect Bitcoin prices in the short run but lose explanatory power over longer horizons, indicating that Bitcoin's link with traditional markets varies across frequencies. This frequency dependence directly motivates a mixed-frequency framework that separates short-run market reactions from long-run macro-financial effects.

Later evidence, however, questioned whether low correlation is enough to make Bitcoin a usable hedge. Smales (2019) shows that Bitcoin's realized volatility, bid–ask spreads, and price-impact costs are far higher than those of traditional safe-haven assets such as gold and government bonds. This is important because hedging effectiveness depends on both correlation and implementation costs. Therefore, a weak correlation may look attractive statistically, but if the hedge is too volatile or too costly to rebalance, much of the apparent variance-reduction benefit disappears in practice (Smales, 2019; Petukhina et al., 2021).

Beyond these cost concerns, crisis-period evidence further weakened Bitcoin's safe-haven case. For example, Conlon and McGee (2020) found that during the COVID-19 crash of March 2020, Bitcoin fell alongside global equities rather than protecting investors. More importantly, their portfolio-level analysis showed that adding Bitcoin increased the Value-at-Risk of a diversified portfolio, reversing the expected hedging effect and deepening losses for investors who held it for protection (Kumar, Mohan & Niveditha, 2025). Similarly, Mariana et al. (2021) found that Ethereum also failed as a safe haven during the same crisis, suggesting that this weakness was not unique to Bitcoin but reflected broader cryptocurrency-market stress. Moreover, Akhtaruzzaman et al. (2021) documented a sharp and persistent rise in Bitcoin–equity correlations during the pandemic. According to Bouri et al. (2024), Bitcoin's diversification benefits in calmer periods may reflect low-uncertainty conditions rather than a stable safe-haven property, meaning that its protective role weakens when macroeconomic uncertainty intensifies.

The conflicting results in the literature point to a fundamental methodological issue: many existing studies rely on simple models—such as the Constant Conditional Correlation model (CCC-GARCH; Bollerslev, 1990), which fixes correlations entirely, or the standard Dynamic Conditional Correlation model (DCC-GARCH; Engle, 2002), which allows correlations to move gradually but through a smooth, parameter-stable recursion—that assume Bitcoin's correlations with traditional assets are either constant or change very smoothly over time. These models often fail to capture sudden structural shifts caused by major changes in the macroeconomic policy environment (Ang & Timmermann, 2012; Conlon & McGee, 2020). As recent research has demonstrated in commodity and cryptocurrency markets (Fang et al., 2019; Ji et al., 2020; Xiong et al., 2024), the true relationship between assets—including how policy uncertainty affects their movements—can change significantly from one economic phase to another. Treating such regime-dependent dependence as a single, time-invariant relationship therefore risks averaging across economically distinct states, destroying valuable timing and leading signals, and producing biased hedge ratios precisely when accurate risk measurement matters most (Jabarabadi, 2024).

1.2.2 Economic policy uncertainty as a systemic risk factor

Economic policy uncertainty has become a central concept in macro-financial research because it captures a form of risk that is distinct from standard market volatility (Brogaard & Detzel, 2015). To measure this risk, Baker et al. (2016) developed the Global Economic Policy Uncertainty (GEPU) index — a GDP-weighted average of national EPU indices across major economies, published at monthly frequency. It therefore captures sustained, structural uncertainty about the policy environment rather than day-to-day market sentiment. This distinguishes it sharply from the VIX, while the VIX reflects short-term implied volatility over a 30-day horizon, GEPU measures a broader, slower-moving risk stemming from long-term ambiguity about government spending, interest rates, trade policy, and regulation (Białkowski, Dang & Wei, 2022).

The theoretical foundation explaining how policy uncertainty influences asset prices was established by Pástor and Veronesi (2012) through a general equilibrium model. In their framework, changes in government policy affect corporate profitability, but investors do not know the full impact immediately. Instead, they update their expectations gradually as new information arrives, which means policy shocks can continue to influence prices and volatility even after the initial announcement. The model therefore identifies two main transmission channels: a direct channel, where policy uncertainty increases discount rates and lowers equity valuations, and an indirect learning channel, where Bayesian updating about future policy trajectories generates persistent volatility beyond the initial shock (Kelly et al., 2016). More importantly, a key prediction of this model is that policy uncertainty acts as a common risk factor. Because all assets are exposed to the same aggregate policy environment, changes in uncertainty simultaneously affect both volatility and cross-asset correlations (Umar et al., 2022; Nguyen & Thanh, 2025).

Moving from theory to evidence, this prediction has been extensively confirmed by empirical data across major asset classes. In equity markets, Liu and Zhang (2015) demonstrated that economic policy uncertainty (EPU) strongly predicts future stock volatility. Crucially, they found that EPU primarily drives the slow-moving, low-frequency component of market risk rather than short-term fluctuations. Beyond equities, this systemic disruption also extends to the commodity sector. For instance, Wang et al. (2015) documented that sudden spikes in EPU elevate crude oil volatility and distort the normal correlation between oil and stock returns, providing evidence that policy uncertainty alters how different markets interact. Similarly, fixed-income markets are highly vulnerable to policy shocks; specifically, Bordo et al. (2016) showed that elevated uncertainty leads to wider credit spreads and amplified bond volatility. This adverse effect intensifies significantly when the credibility of monetary policy is questioned (Gulen & Ion, 2016). Together, this evidence confirms that policy uncertainty drives individual asset risk and reshapes correlations across the global investment landscape (Umar et al., 2022).

While these effects are well-documented in traditional markets, how policy uncertainty transmits to cryptocurrency markets is a more recent and far less settled question. Because Bitcoin appears to behave independently of economic and financial developments (Baur et al., 2018; Corbet et al., 2018), an early strand argued it could act as a hedge in its own right (Urquhart & Zhang, 2019; Platanakis & Urquhart, 2020). Panagiotidis et al. (2019) qualified this, finding macro-financial development — measured by the Dow Jones Index, exchange rates, and crude oil prices — to significantly drive Bitcoin prices in the short run while becoming insignificant in the long run. Focusing on policy uncertainty, Demir et al. (2018) provided the first formal evidence that EPU predicts Bitcoin returns, arguing that rising uncertainty erodes investor trust in the global financial system and conventional currencies and thereby raises Bitcoin's attractiveness — and urging policymakers and investors to watch the effect of policy uncertainty on Bitcoin returns.

According to Fang et al. (2019), who applied GARCH-MIDAS and DCC-MIDAS models with the GEPU index to examine Bitcoin's volatility and hedging capacity, the results confirmed that GEPU significantly shapes Bitcoin's long-run volatility and its co-movement with conventional assets, and that incorporating this macro signal modestly improves hedging performance. Together with the broader EPU literature, this establishes two points clearly: policy uncertainty is a genuine, measurable driver of long-run asset dynamics, and the GARCH-MIDAS framework is the appropriate tool for capturing it (Dai et al., 2022). What remains unresolved is whether this GEPU effect remains stable across different uncertainty regimes, because Fang et al. (2019) implicitly assume a constant uncertainty–Bitcoin relationship, while more recent evidence suggests that cryptocurrency responses to policy uncertainty shift across structurally different macroeconomic environments, making a structural-break extension necessary (Jabarabadi, 2024; Oldani, 2024).

1.3 Research purpose

The primary objective of this study is to evaluate whether conditioning dynamic hedge ratios on global economic policy uncertainty (GEPU) improves Bitcoin's ability to reduce portfolio risk across three conventional asset classes — global equities, commodities, and investment-grade bonds. To this end, the thesis develops a GARCH-MIDAS-X and DCC-MIDAS-X framework augmented with Bai–Perron structural breaks, and assesses whether the resulting hedge ratios deliver superior portfolio variance reduction relative to standard benchmarks, both before and after accounting for transaction costs.

Specifically, this thesis aims to accomplish the following three core purposes:

- **To quantify the impact of GEPU on long-run volatility:** Using the GARCH-MIDAS-X specification, this study examines whether policy uncertainty explains the persistent volatility of Bitcoin and the three conventional assets beyond the information already contained in daily realized variance. It also tests whether this GEPU–volatility transmission changes across the uncertainty regimes identified by the endogenous Bai–Perron procedure.

- **To analyze the impact of GEPU on long-run conditional correlations:** Using the DCC-MIDAS-X framework, this study investigates the direction, magnitude, and time-varying nature of GEPU's effect on the long-run conditional correlations between Bitcoin and each conventional asset index. This analysis explores how cross-market co-movements evolve and adapt across qualitatively distinct macroeconomic environments.

- **To evaluate Bitcoin's hedging performance in practical settings:** This study assesses whether GEPU-conditioned, regime-aware optimal hedge ratios strengthen Bitcoin's capacity to reduce portfolio variance, delivering statistically and economically significant improvements over traditional benchmark models. To establish whether this hedging capacity holds up for institutional investors in practice, the evaluation combines in-sample performance, out-of-sample rolling-window forecasting, and transaction-cost-adjusted returns at realistic cost levels.

1.4 Research objects and scope

1.4.1 Research objects

To achieve the primary research purpose, this study focuses on four specific objectives: First, to estimate the GARCH-MIDAS-X model for four distinct assets (Bitcoin, MSCI World Index, S&P GSCI, and PIMCO Bond Index) to determine if monthly Global Economic Policy Uncertainty (GEPU) is a statistically significant driver of long-run asset volatility. Second, to estimate bivariate DCC-MIDAS-X models for three market pairs (Bitcoin–MSCI, Bitcoin–S&P GSCI, and Bitcoin–PIMCO) to quantify the direct impact of monthly GEPU shocks on their long-run dynamic correlations. Third, to apply the endogenous Bai–Perron procedure to detect major macroeconomic structural breaks within the sample period and evaluate if incorporating these breaks improves overall model fit. Fourth, to construct optimal time-varying hedge ratios from the estimated covariance matrices and assess how effectively Bitcoin hedges portfolio risk under the GEPU-augmented, regime-aware DCC-MIDAS-X model relative to the benchmark DCC-MIDAS model, evaluated across both in-sample and out-of-sample windows, with and without daily transaction costs. Building on these results, the study finally aims to translate its findings into practical hedging strategies, offering investors concrete guidance on how to adjust Bitcoin-based hedges across different asset classes and macroeconomic regimes.

1.4.2 Research scope

The study will examine a dataset consisting of 2,296 daily asset return observations and 110 monthly Global Economic Policy Uncertainty (GEPU) observations, spanning from October 2016 to December 2025. This nine-year sample window is intentionally selected to cover four major policy uncertainty regimes: the U.S.–China trade war (2018–2019), the COVID-19 pandemic (2020–2021), the global monetary tightening cycle (2022–2023), and the spot Bitcoin ETF approval period (2024–2025). The asset universe focuses on Bitcoin as the hedging instrument paired against three conventional asset class proxies: the MSCI World Index for global equities, the S&P GSCI for commodities, and the PIMCO Investment Grade Corporate Bond Index for fixed income. The GEPU index serves as the sole low-frequency macroeconomic conditioning variable to isolate the policy uncertainty transmission channel. Methodologically, the econometric scope comprises four nested model specifications estimated across the three asset pairs, resulting in 16 univariate volatility models and 12 bivariate correlation models to systematically decompose the risk-reduction contributions of the GEPU channel, structural breaks, and their joint interaction.

1.5 Research questions

The research questions are designed to systematically evaluate how macroeconomic uncertainty transmits into Bitcoin's risk dynamics and what this means for managing portfolio risk with Bitcoin. By progressing logically from Bitcoin's individual volatility to its correlations with major asset classes, and finally to its practical hedging applications, this study addresses three core questions:

RQ1: Does monthly Global Economic Policy Uncertainty (GEPU) significantly drive the long-run volatility components of Bitcoin, the MSCI World Index, the S&P GSCI, and the PIMCO Bond Index, beyond their short-term daily fluctuations?

RQ2: Does monthly GEPU significantly influence the long-run conditional correlations across the three distinct market pairs (Bitcoin–MSCI, Bitcoin–S&P GSCI, and Bitcoin–PIMCO), and do these cross-asset relationships structurally shift across different macroeconomic regimes?

RQ3: Does the GEPU-augmented, regime-aware DCC-MIDAS-X framework provide statistically, and economically superior Bitcoin’s hedging effectiveness compared to the standard DCC-MIDAS baseline, particularly when evaluated across out-of-sample forecasting windows and adjusted for daily transaction costs?

1.6 Research method and data

This study uses a quantitative approach to model Bitcoin's volatility and assess its hedging effectiveness against three conventional asset classes under global economic policy uncertainty. The empirical design follows a three-stage framework based on the GARCH-MIDAS and DCC-MIDAS tradition, combining mixed-frequency volatility decomposition, dynamic correlation modeling, and out-of-sample portfolio evaluation. This framework links daily asset returns with monthly GEPU data, separates short-run market noise from long-run macro-driven risk, and allows the GEPU–risk relationship to vary across structural regimes.

First, a univariate GARCH-MIDAS-X model decomposes each asset's conditional variance into a daily GARCH(1,1) component and a monthly long-run component driven by realized volatility and GEPU through a MIDAS beta weighting scheme. In addition, structural-break dummies, identified using the Bai and Perron (2003) sequential sup-F procedure, are interacted with the MIDAS coefficients to capture regime-specific uncertainty effects.

Second, a bivariate DCC-MIDAS-X model estimates the dynamic correlation between Bitcoin and each conventional asset. Specifically, short-run correlations follow a GARCH-type recursion, while long-run correlations are modeled using MIDAS-weighted realized correlations and lagged GEPU. To ensure valid correlation estimates, a Fisher-z transformation is applied. Moreover, structural-break dummies are included to allow regime shifts in the GEPU–correlation relationship.

Third, the estimated conditional covariance matrices are then used to construct optimal time-varying hedge ratios that minimize hedged portfolio variance. Accordingly, hedging effectiveness is measured as the proportional variance reduction relative to an unhedged position across model specifications and asset pairs.

Model parameters are estimated using two-stage quasi-maximum likelihood following Colacito et al. (2011). Meanwhile, robust inference is based on Bollerslev and Wooldridge's (1992) sandwich covariance matrix. Forecast and hedging improvements are subsequently tested using the Clark and West (2007) statistic, supplemented by the Diebold and Mariano (1995) test with Newey–West standard errors. Finally, robustness checks include rolling-window out-of-sample evaluation using a 70/30 split, sensitivity tests with 60/40 and 80/20 splits, transaction-cost adjustments of 10, 25, and 50 basis points, and break-date perturbation by ±3 months.

Data collection

This study combines high- and low-frequency data from October 2016 to December 2025. At the daily level, the analysis uses Bitcoin and three conventional asset-class proxies — global equities (MSCI World), commodities (S&P GSCI), and bonds (PIMCO) — from which returns and realized measures are constructed to feed the MIDAS components. Bitcoin price data are obtained from the CoinDesk Bitcoin Price Index (XBX), a USD-denominated reference rate for the spot price of Bitcoin constructed from real-time prices across multiple constituent exchanges. At the monthly level, the Global Economic Policy Uncertainty (GEPU) index serves as the macroeconomic conditioning variable. The sample deliberately spans four distinct regimes — the trade war, the COVID-19 pandemic, the 2022–2024 tightening cycle, and the post-ETF institutionalization phase — so that the structural-break analysis captures the full range of uncertainty environments relevant to Bitcoin's hedging role.

1.7 Contribution and significance of the research

This thesis contributes to the literature on cryptocurrency hedging by examining how global economic policy uncertainty and structural breaks jointly affect Bitcoin's risk-reduction role across conventional asset classes. Specifically, it focuses on the transmission mechanism of Global Economic Policy Uncertainty (GEPU) and the possibility that Bitcoin's volatility, correlations, and hedging capacity change across macroeconomic regimes. In doing so, the study provides new evidence on whether Bitcoin can function as a practical hedging instrument in financial markets, not only as a speculative crypto asset, but also as a regime-dependent tool for portfolio risk reduction. By addressing key gaps in the existing literature and introducing a regime-aware mixed-frequency framework, this research makes several important contributions.

First, this study makes a methodological contribution by integrating endogenous structural-break detection into a GARCH-MIDAS-X and DCC-MIDAS-X model framework for evaluating Bitcoin's hedging effectiveness. While Fang et al. (2019) apply GARCH-MIDAS and DCC-MIDAS models with GEPU to Bitcoin, their framework assumes that the uncertainty–risk relationship remains stable over time. Meanwhile, Xiong et al. (2024) incorporate structural breaks into a DCC-MIDAS-X setting, but their analysis does not focus on cryptocurrency hedging. This thesis bridges these two strands by combining Bai–Perron structural breaks with mixed-frequency volatility and correlation models, allowing GEPU's transmission into Bitcoin's risk dynamics to vary across distinct macroeconomic regimes.

Second, this research contributes to the empirical literature by extending the analysis of Bitcoin's hedging role to a more recent and institutionally relevant sample period. The study covers 2016–2025, a period that includes the COVID-19 crisis, the 2022–2024 monetary tightening cycle, and the post-spot-Bitcoin-ETF environment. This updated sample allows the research to examine Bitcoin's hedging capacity under multiple uncertainty regimes that are not fully captured in earlier studies. It also broadens the asset scope by evaluating Bitcoin against global equities, commodities, and investment-grade bonds, rather than focusing on a single asset pair.

Third, this thesis contributes practically by developing a regime-specific hedging framework that can inform portfolio risk management. Instead of treating Bitcoin's hedging role as fixed, the study provides a structure for estimating time-varying hedge ratios across different asset classes and uncertainty regimes. This is useful for portfolio managers because it links hedging decisions not only to daily market movements, but also to slower-moving macroeconomic uncertainty. The inclusion of out-of-sample evaluation and transaction-cost adjustment further strengthens the practical relevance of the framework.

The significance of this research lies in both its academic and practical implications.

- **Academic Contribution:** This thesis extends the cryptocurrency hedging literature by introducing a macro-conditioned and regime-aware approach to Bitcoin risk modelling. It relaxes the constant-parameter assumption used in much of the earlier literature and provides a framework for studying Bitcoin's volatility and co-movement structure across structural regimes. By combining GEPU conditioning, Bai–Perron structural breaks, GARCH-MIDAS-X, and DCC-MIDAS-X, the study offers a more comprehensive foundation for future research on cryptocurrency hedging, macro-financial uncertainty, and mixed-frequency volatility modelling.
- **Practical Implications:** For investors and portfolio managers, this research provides a framework for assessing when Bitcoin may serve as a useful hedging instrument across equities, commodities, and bonds. Its regime-specific design helps practitioners evaluate Bitcoin's risk-reduction role under changing macroeconomic conditions rather than relying on a single average relationship. This makes the study relevant for institutional investors seeking to manage portfolio risk during periods of recurring systemic uncertainty.

In conclusion, this thesis reframes the Bitcoin hedging debate from whether Bitcoin serves as a hedge in general to when, across which regimes, and under what conditions it can reduce portfolio risk. In doing so, the research connects mixed-frequency volatility modelling with real-world portfolio risk management in institutional practice under changing macroeconomic uncertainty regimes.

1.8 Structure of the research

In addition to the introduction, conclusion, list of acronyms, list of tables, figures, list of references and appendices, the paper is structured in five chapters as follows:

Chapter 1: INTRODUCTION

Chapter 2: LITERATURE REVIEW AND HYPOTHESIS DEVELOPMENT

Chapter 3: METHODOLOGY AND DATA COLLECTION

Chapter 4: EMPIRICAL RESULTS

Chapter 5: CONCLUSIONS AND RECOMMENDATIONS

Summary of Chapter 1

Chapter 1 establishes the foundation of the thesis by starting from a practical portfolio problem, as global recent crises have exposed the weakness of traditional hedges and intensified the search for alternative assets capable of reducing portfolio drawdowns when conventional markets fall together. Against this backdrop, the chapter explains why Bitcoin has become relevant to institutional risk management, given its transition from a speculative digital asset to an increasingly recognized portfolio instrument. It then situates the research within the conflicting and sample-dependent evidence on Bitcoin's hedging role, showing that the same asset has been classified as a hedge, safe haven, or diversifier depending on the period and market conditions examined. This inconsistency is traced to a key modelling limitation, since many existing frameworks assume a stable relationship between uncertainty and asset co-movements, even though recent macroeconomic shocks suggest that this relationship changes across regimes. Building on this diagnosis, the chapter defines the research purpose, scope, and objectives, while emphasizing the need for a model that can capture both mixed-frequency macroeconomic effects and structural shifts in the post-2018 environment. It then outlines a methodology that combines GARCH-MIDAS-X and DCC-MIDAS-X modelling with endogenous Bai–Perron structural breaks, estimated across 16 univariate volatility models and 12 bivariate correlation models for Bitcoin paired with equities, commodities, and bonds. Finally, the chapter highlights the contribution of the thesis by proposing a regime-aware and macro-conditioned hedging framework designed to assess whether Bitcoin can improve portfolio risk reduction under global economic policy uncertainty and provide practical guidance for institutional portfolio managers.

CHAPTER 2: LITERATURE REVIEW AND HYPOTHESIS DEVELOPMENT

2.1 Literature review

2.1.1 Portfolio Hedging Theory: From static to dynamic frameworks

Hedging in financial markets traces its intellectual roots to portfolio theory, where Markowitz (1952) first showed that pairing a risky asset with another whose returns are less than perfectly correlated lowers overall portfolio variance without proportionally sacrificing expected return. The size of that reduction hinges entirely on the covariance between the two assets, a deceptively simple insight that reshaped how practitioners think about risk by locating it in how assets move together rather than in any asset alone. Sharpe (1964) and Lintner (1965) carried this logic into the Capital Asset Pricing Model, drawing the now-fundamental line between systematic risk, the part of an asset's variance that moves with the market and cannot be diversified away, and idiosyncratic risk, which diversification eliminates at no cost. An asset with low or negative market beta offers genuine insurance against systematic shocks regardless of its own expected return, a prediction borne out across international equity markets by Fama and MacBeth (1973) and ultimately formalized into the framework for evaluating hedging instruments by Baur and Lucey (2010).

Operationalizing this insight, Johnson (1960) translated the Markowitz variance-minimization principle into a concrete hedging formula. If an investor holds a spot position in asset S and takes an offsetting position in a futures contract F, the variance of the combined portfolio is:

$$Var(portfolio) = \sigma_S^2 + \beta^2 \cdot \sigma_F^2 - 2\beta \cdot Cov(r_S, r_F)$$

Minimizing this expression with respect to β yields the variance-minimizing hedge ratio, the position in the hedging instrument that maximizes portfolio variance reduction:

$$\beta^* = Cov(r_S, r_F)/Var(r_F) = \rho_{SF} \cdot (\sigma_S/\sigma_F)$$

where ρ_{SF} is the conditional correlation and σ_S, σ_F are the conditional standard deviations of the spot and futures returns. Ederington (1979) then operationalized the quality of this hedge through the hedging effectiveness (HE) index, defined as the proportional reduction in portfolio variance relative to the unhedged baseline:

$$HE = \frac{[Var(r_S) - Var(r_S - \beta^* r_F)]}{Var(r_S)}$$

An HE value of one indicates perfect hedging, while a value of zero indicates no variance reduction. Together, the optimal hedge ratio β* and the HE index form the empirical foundation of portfolio hedging research because they can evaluate hedging performance across very different instruments without changing the underlying logic. As Ederington (1979) noted, the framework does not depend on the nature of the hedging asset, which explains why it can be applied from agricultural futures (Johnson, 1960) and equity index futures (Ku et al., 2007) to currency forwards (Chang et al., 2011) and Bitcoin.

A critical limitation of this static OLS implementation, however, is its assumption that the optimal hedge ratio is constant over time. Since β^* depends on time-varying volatility and correlation, a static estimate will be systematically misspecified whenever the covariance structure of returns shifts (Kroner & Sultan, 1993). Consequently, Baillie and Myers (1991) showed that static hedge ratios become biased when markets are unstable, leaving investors exposed at the exact moment hedging accuracy matters most.

The key methodological advance was the ARCH model of Engle (1982), later generalized by Bollerslev (1986) into the GARCH(1,1) specification:

$$\varepsilon_t = z_t \cdot \sqrt{(h_t)}, \quad z_t \sim N(0,1)$$

$$h_t = \omega + \alpha\varepsilon_{t-1}^2 + \beta h_{t-1}$$

where h_t is the conditional variance, $\omega > 0$, $\alpha \geq 0$, $\beta \geq 0$, and $\alpha + \beta < 1$ ensures stationarity. The value of this model lies in its ability to capture volatility clustering, where large shocks tend to be followed by further large shocks and calm periods by further calm periods. This is essential for hedging because the optimal hedge ratio $\beta^* = \rho \cdot (\sigma_S/\sigma_F)$ depends on time-varying volatility, GARCH therefore turns hedging from a static allocation problem into a dynamic risk-management process, in which hedge ratios must adjust as market volatility changes.

Naturally, extending GARCH to the bivariate setting required modeling the joint conditional distribution of two asset returns, including their time-varying covariance. For example, Bollerslev (1990) introduced the Constant Conditional Correlation (CCC) model, which constrained the conditional correlation matrix to be fixed at its unconditional sample value. While computationally tractable, Longin and Solnik (2001) demonstrated that this assumption is empirically false: correlations between international equity markets increase substantially during joint market stress, meaning that CCC models systematically understate co-movement risk precisely when it is most costly. Engle (2002) resolved this with the Dynamic Conditional Correlation (DCC) model, in which each asset's return is first standardized by its GARCH conditional standard deviation and then the correlations among standardized residuals evolve through a GARCH-type recursion, allowing correlations to rise during episodes of joint volatility and revert toward their long-run mean during calmer periods. Reassuringly, Ku, Chen, and Chen (2007) confirmed empirically that DCC-based hedge ratios dominate static and CCC-based alternatives in out-of-sample variance reduction across equity, foreign exchange, and commodity markets.

Despite this progress, the standard DCC model has one important limitation for macro-financial analysis. It treats the long-run target correlation, or the level to which short-run correlations revert, as a fixed full-sample parameter (Engle, 2002; Colacito et al., 2011). As a result, it cannot clearly distinguish temporary correlation movements caused by daily market noise from persistent shifts driven by macroeconomic forces. Colacito, Engle, and Ghysels (2011) address this limitation by proposing a mixed-frequency architecture in which the long-run correlation component can be linked to low-frequency macroeconomic variables.

The same logic appears in the GARCH-MIDAS model of Engle et al. (2013), which decomposes return volatility into short-run and long-run components while allowing variables observed at different frequencies to enter the model. DCC-MIDAS then extends this idea to correlations, allowing long-run co-movement to vary with slowly evolving economic conditions. Together, GARCH-MIDAS and DCC-MIDAS overcome a key weakness of single-frequency multivariate models such as BEKK (Engle & Kroner, 1995). Unlike BEKK, which requires variables to be observed at the same frequency, the MIDAS structure integrates low-frequency macroeconomic variables into high-frequency volatility and correlation processes without aggregating daily data or losing information (Engle et al., 2013).

2.1.2 Bitcoin as a portfolio instrument

2.1.2.1 Diversifier, hedge and safe haven

Evaluating any asset's role in a portfolio requires a precise taxonomy, and the foundational tripartite classification that has since become the standard terminology of the Bitcoin hedging literature was established by Baur and Lucey (2010) and Baur and McDermott (2014). Under this framework, a hedge is an asset that remains uncorrelated or negatively correlated with a reference portfolio on average across market conditions, while a safe haven provides the same protective correlation specifically during periods of extreme market stress. A diversifier, by contrast, has a low but positive average correlation with the reference asset, allowing it to reduce portfolio variance in normal conditions but offering weaker protection when markets experience joint downside movements.

Equally important, a conceptual distinction must be drawn between two uses of the term hedging. Direct hedging neutralizes the price risk of an asset through an offsetting position in a closely linked instrument, typically its own futures contract, whose near-perfect co-movement with the spot asset eliminates almost all exposure (Johnson, 1960; Ederington, 1979). On the other hand, cross-asset hedging serves a fundamentally different purpose. Rather than neutralizing a single exposure, it lowers the variance of a multi-asset portfolio by combining assets whose returns are weakly or negatively correlated (Markowitz, 1952; Baur & Lucey, 2010). In this setting an instrument's hedging value arises precisely from its low co-movement with the position being protected, so an asset that moves in lockstep with the underlying, such as its own futures, offers no diversification benefit and cannot reduce portfolio variance. It is for this reason that Bitcoin's hedging role is meaningfully assessed against equities, commodities, and fixed income rather than against Bitcoin-linked derivatives, as its value as a portfolio hedge depends entirely on whether its correlation with conventional asset classes remains low or negative across market regimes (Singh et al., 2024; Bouazizi et al., 2025).

These distinctions carry profound practical consequences for hedging strategy. A diversifier lowers portfolio volatility during calm markets through imperfect co-movement, but when correlations converge during systemic crises — as the empirical literature consistently documents they do (Longin & Solnik, 2001; Ang et al., 2006) — it fails to provide protection at precisely the worst moment. A safe haven, by contrast, provides insurance that is most active during crisis episodes, functioning as a put option on the portfolio. Translating these concepts into the Johnson–Ederington hedging framework: a hedge satisfies $\beta^* \leq 0$ unconditionally; a safe haven satisfies $\beta^* \leq 0$ conditionally on extreme market stress; a diversifier has $0 < \beta^* < 1$. Under any of these configurations, including the asset in a portfolio reduces the variance of the hedged position relative to the unhedged benchmark, but the magnitude and reliability of that reduction depend critically on whether the correlation regime is stable or time-varying (Colacito et al., 2011).

Given this background, Bitcoin's case as a diversifier rests on its structural separation from the traditional financial system. Nakamoto (2008) capped Bitcoin's supply at 21 million coins and designed it as a decentralized ledger outside the control of central banks and governments. As a result, Bitcoin is less directly exposed to monetary and fiscal policy than conventional assets. Its fixed supply also mirrors the scarcity logic behind gold's store-of-value role (Barsky & Summers, 1988; Selmi et al., 2018). When confidence in fiat currencies weakens, an asset with a hard supply limit and no central issuer may become more attractive, creating the low or negative correlation needed for hedging. This is consistent with the flight-to-quality mechanism, where investors shift toward assets perceived as safer or less connected to the source of stress (Caballero & Krishnamurthy, 2008). According to Demir et al. (2018), rising policy uncertainty can increase Bitcoin's attractiveness by weakening confidence in conventional financial systems, suggesting that capital may flow into

Bitcoin when investors perceive it as an alternative asset outside the traditional financial architecture.

Against this theoretical optimism stands a compelling counter-argument: Bitcoin is an extraordinarily volatile asset. Smales (2019) quantified the gap precisely — Bitcoin's realized volatility runs roughly 5–8× that of gold and 15–20× that of US government bonds across comparable sample periods, an order-of-magnitude difference that no correlation advantage can offset. Under the variance-minimizing framework, this implies that even a moderately negative correlation will yield only a small reduction in the variance of the hedged portfolio, because $\beta^* = \rho \cdot (\sigma_{ref}/\sigma_{Bitcoin})$ will be very small in absolute value. This extreme volatility means that Bitcoin's practical hedging capacity is structurally limited regardless of its correlation properties (Conlon and McGee, 2020).

2.1.2.2 Empirical evidence on Bitcoin's hedging properties

The early empirical literature was generally optimistic about Bitcoin's hedging and diversification role, with different methods pointing to a similar conclusion. Using univariate GARCH models, Dyhrberg (2016) showed that Bitcoin could hedge both the FTSE 100 and the US dollar while Bouri et al. (2017) applied a DCC-GARCH model with quantile dummies and found that Bitcoin could diversify extreme downside movements in equity returns. Similarly, Baur et al. (2018) found that Bitcoin was weakly correlated with equities in both normal and stress periods, although they emphasized that it was used mainly as a speculative investment rather than as an alternative currency or medium of exchange. Multivariate studies reached similar findings, Guesmi et al. (2018) showed that adding Bitcoin to a portfolio of emerging-market equities, gold, and crude oil reduced portfolio risk, while Ji et al. (2018) and Corbet et al. (2018) reported comparable hedging benefits using graph-theoretic variance decomposition and time-frequency connectedness methods. However, this optimistic view reflected an earlier period when Bitcoin was still largely retail-driven and relatively separate from the traditional financial system. As a result, its low unconditional correlation may have reflected the market conditions of the period rather than an inherent property of Bitcoin. This is consistent with Brière et al. (2015), who found that a small 3–5% Bitcoin allocation improved risk-adjusted returns during a period when Bitcoin was still relatively detached from traditional financial markets.

A critical limitation of these early findings, visible even at the time, was their reliance on unconditional correlation estimates from relatively stable sample periods. This approach is problematic because, as the Baur and Lucey (2010) framework makes clear, it cannot separate a genuine hedge, which requires negative average correlation, from a fair-weather diversifier, which only shows low correlation in calm markets. In practical terms, this distinction matters because a portfolio manager who builds a hedge from an unconditional correlation estimated during a calm regime implicitly assumes that the same relationship will persist into the next regime, yet the Markov-switching literature on financial correlations systematically rejects this assumption (Hamilton, 1989; Ang & Bekaert, 2002). Moreover, the frequency dimension creates another complication, as Bouri et al. (2022) showed that Bitcoin's hedging capacity operates at weekly but not daily horizons, providing an early signal that the Bitcoin–equity relationship changes across investment frequencies and that daily-frequency models may miss the low-frequency co-movement structure where macro-driven correlations emerge.

According to Ang and Timmermann (2012), asset correlations can shift sharply across market regimes, especially during volatile periods. This helps explain why Bitcoin's early correlation estimates became unstable as longer samples began to cover multiple market cycles. Bitcoin's near-zero equity correlation in 2010–2017 largely reflected its retail-dominated and relatively detached market structure, a relationship that weakened as the asset became more financialized (Corbet et al., 2020). Baur, Hong, and Lee (2018) documented this instability, showing that Bitcoin's equity correlation ranged from −0.15 to +0.35 across sample windows. Such variation is difficult to treat as a stable structural relationship. Instead, it suggests regime-switching dynamics, where full-sample correlations average together very different regime-specific relationships and may misrepresent Bitcoin's true hedging role (Wang et al., 2025).

The deeper structural problem was exposed during episodes of joint market stress, where Bitcoin's behavior diverged sharply from theoretical hedge predictions. Klein, Pham Thu, and Walther (2018) demonstrated through multivariate GARCH specifications that unlike physical gold — whose negative equity-beta during downturns reflects the flight-to-quality mechanism of Caballero and Krishnamurthy (2008) — Bitcoin's conditional beta with equities turns positive and significant during the worst 5% of equity return days, amplifying rather than offsetting portfolio losses. This inverse safe-haven behavior has a clear theoretical interpretation: during acute systemic stress, the margin-call and liquidity-demand channels dominate, forcing leveraged investors to liquidate all liquid positions simultaneously regardless of asset class, generating synchronized drawdowns that overwhelm any diversification benefit built on calm-period correlations. The COVID-19 crash of March 2020 provided a decisive natural experiment: Bitcoin's price fell by more than 25% within a single week while its conditional correlation with the S&P 500 rose from approximately 0.15 to above 0.60 (Conlon and McGee, 2020) — a spike of roughly 4 standard deviations relative to the pre-crisis baseline, confirming that Bitcoin's hedging capacity is not merely time-varying but collapses precisely during the episodes when portfolio protection is most valuable.

2.1.2.3 Regime-dependence and the asset-class heterogeneity

According to Bouoiyour et al. (2022) and Apergis et al. (2023), across all asset class pairs, the most consistent empirical finding is that Bitcoin’s hedging, safe-haven, and diversification properties are not fixed characteristics but are regime-dependent: they vary systematically across macroeconomic environments, investment horizons, and types of market shock. For equity markets, Bouri, Roubaud, and Kristoufek (2019) used a cross-quantilogram approach to show that Bitcoin’s dependence with global equity indices is concentrated in the upper tail of the joint distribution — Bitcoin co-moves with rising equity markets but exhibits weaker dependence during bear markets, a pattern more consistent with a cyclical risk asset than a hedge. For commodity markets, Selmi et al. (2018) documented that Bitcoin can act as a hedge, safe haven, or diversifier for crude oil depending on the prevailing uncertainty regime, with safe-haven properties present primarily during episodes of extreme oil price volatility. For bond markets, Fang et al. (2019) found a significantly negative GEPU effect on the Bitcoin–bond correlation, implying that rising policy uncertainty causes Bitcoin and investment-grade bonds to move in opposite directions, strengthening Bitcoin's diversification value for fixed-income portfolios.

This cross-asset heterogeneity in the sign and magnitude of Bitcoin's macro-conditioned correlations has a crucial methodological implication. If Bitcoin's correlation with each asset class depends on the macroeconomic regime, then any model that estimates a single unconditional correlation is misspecified by construction (Raza et al., 2023). Standard DCC models that allow short-run correlation fluctuations without conditioning on macroeconomic variables will systematically produce incorrect hedge ratios during regime transitions (Antonakakis et al., 2019). Correctly measuring Bitcoin's hedging value therefore requires an econometric framework that explicitly models the role of macroeconomic uncertainty in driving long-run correlation dynamics. Such a framework decomposes the correlation structure into short-run and long-run components and directly links the long-run target to macroeconomic conditioning variables, allowing both the magnitude and sign of the macro effect to adapt to the prevailing uncertainty regime (Xia et al., 2023; Aras et al., 2025).

2.1.3 Global Policy Uncertainty and financial risk

The systematic measurement of economic policy uncertainty as a quantifiable financial variable was established by Baker, Bloom, and Davis (2016), whose EPU index resolved the longstanding difficulty of converting a fundamentally subjective macroeconomic concept into a time series amenable to empirical analysis. Their index aggregates three components: the frequency of newspaper articles in major national outlets referencing economic uncertainty alongside policy-relevant terms; the dollar value of expiring federal tax code provisions, capturing enacted but impermanent fiscal policy changes; and the dispersion among professional economic forecasters for key macroeconomic variables, reflecting genuine ambiguity about the future policy environment. The GDP-weighted global extension — the GEPU index of Davis (2016) — applies this methodology across 21 major economies, collectively accounting for approximately 67% of global output on a PPP-adjusted basis and roughly 75% at market exchange rates. The monthly GEPU value is computed as the GDP-weighted average of national EPU indices, expressed as:

$$GEPU_t = \sum_{i=1}^{N} w_i \cdot EPU_{i,t}$$

Where:

- $GEPU_t$ is the global economic policy uncertainty index in month t;
- N is the number of economies included in the index, equal to 21 in the Davis (2016) global extension;
- $EPU_{i,t}$ is the renormalized policy uncertainty index for country i in month t;
- w_i is country i's GDP weight in month t, calculated as its share of total PPP-adjusted GDP among the countries included in the index.

This weighting structure ensures GEPU reflects genuine global policy uncertainty rather than the idiosyncratic policy cycle of any single economy.

Crucially, however, a critical conceptual distinction separates GEPU from equity market volatility indices — such as the VIX — that are frequently used as uncertainty proxies. The VIX measures the implied volatility of S&P 500 options over a thirty-day horizon, capturing short-horizon investor fear about near-term price fluctuations (Whaley, 2000). GEPU, by contrast, reflects the slower-moving, persistent ambiguity about the future direction of fiscal, monetary, trade, and regulatory policy — the kind of structural uncertainty that shapes the risk environment for months or years and operates at a fundamentally different frequency than daily market volatility (Baker et al., 2016; Davis, 2016). Because GEPU is observed monthly and affects markets through slow-moving policy channels, its impact cannot be properly captured by models that focus only on daily price movements (Baker et al., 2016; Pástor & Veronesi, 2013). A mixed-frequency framework is therefore necessary because it preserves high-frequency return information while allowing monthly GEPU shocks to enter the long-run volatility and correlation components through a flexible distributed-lag structure (Ghysels et al., 2006; Engle et al., 2013), which prevents the persistent effect of policy uncertainty from being obscured by short-run trading noise.

Pástor and Veronesi (2013) provided the canonical general equilibrium theory linking policy uncertainty to asset prices. In their model, government policy changes affect firms' profitability and investors learn about those changes gradually through Bayesian updating, generating two distinct transmission channels. The first is a discount rate channel, in which uncertainty about future policy raises the risk premium investors demand, compressing equity valuations and widening credit spreads even before any policy change is enacted, because uncertainty is itself a source of systematic risk common to all firms exposed to the same policy environment. The second is a cash flow learning channel, in which Bayesian updating generates persistent return volatility that can outlast the initial shock by quarters as investors revise their beliefs about the terminal policy state. A central prediction is that policy uncertainty acts as a common factor across asset classes: because all assets are exposed to the same aggregate policy environment, GEPU shocks affect not only individual return distributions but also the joint dependence structure between assets, raising cross-asset correlations during high-uncertainty periods and eroding diversification benefits. Ahir, Bloom, and Furceri (2022) validated this property globally, documenting that uncertainty spikes propagate synchronously across countries and asset classes, with spillovers large enough to shift global financial conditions independently of domestic monetary policy.

According to Arouxet et al. (2024), policy uncertainty transmits to asset prices through theoretically distinct, asset-specific channels — each with a predictable sign and frequency structure directly testable within the MIDAS framework. For equity markets, the dominant channel is the discount rate mechanism formalized by Pástor and Veronesi (2023): elevated policy uncertainty raises the risk premium investors demand for holding claims on uncertain corporate cash flows, compressing equity valuations even before any policy change materializes. Liu and Zhang (2015) quantified this channel precisely, showing that a one-standard-deviation increase in EPU predicts a 15–20% increase in the subsequent month's realised equity volatility, with an out-of-sample R^2 of 6.2% — exceeding the predictive power of traditional financial indicators such as the dividend-price ratio and the term spread. Because this channel operates through the revision of discount rates — a process that unfolds over weeks and months as new policy signals accumulate — its empirical footprint is concentrated in the persistent, low-frequency component of equity volatility rather than in daily return shocks (Caporale et al., 2024). Moreover, Brogaard and Detzel (2015) confirmed this frequency signature in the cross-section, estimating that EPU commands a risk premium of approximately 2.1% per annum in US equity returns — roughly one-third of the total equity risk premium — establishing policy uncertainty as a priced systematic factor rather than diversifiable noise.

For commodity markets, the transmission operates through a demand uncertainty channel whereby firms facing ambiguous policy defer investment and procurement, contracting demand for energy and raw materials and generating persistent commodity price volatility (Wang et al., 2015; Shahzad et al., 2017). Wang et al. (2015) documented that EPU Granger-causes crude oil price volatility at the 1% significance level, with a one-standard-deviation EPU shock raising oil volatility by approximately 8–12% over the subsequent quarter — an effect that persists for six to nine months before dissipating, confirming the slow-moving, business-cycle-frequency nature of the transmission. Crucially, Shahzad et al. (2017) demonstrated that this effect is concentrated in the tails of the commodity return distribution: GEPU predicts commodity volatility at the 5th and 95th quantiles with explanatory power roughly three times stronger than at the median — meaning that the GEPU–commodity relationship is primarily a tail phenomenon that manifests during extreme market conditions rather than uniformly across the distribution. This nonlinear, tail-dependent transmission pattern carries a direct methodological implication: a single GEPU slope coefficient estimated across the full distribution will understate the true effect during high-uncertainty regimes and overstate it during calm periods (Liu et al., 2024).

For fixed-income markets, two competing channels operate simultaneously, generating theoretically ambiguous net effects that depend on which mechanism dominates in a given regime. The credit risk channel follows the same logic as the equity discount rate mechanism: rising policy uncertainty widens credit spreads and amplifies bond return volatility, particularly for investment-grade instruments whose valuations are sensitive to the monetary policy outlook. Bordo, Duca, and Koch (2016) estimated that a one-standard-deviation increase in EPU is associated with a 30–50 basis point widening in investment-grade corporate credit spreads and a 10–15% increase in bond return volatility — an economically substantial effect concentrated in the intermediate maturity segment (5–10 years) where duration sensitivity to policy rate expectations is highest. However, the flight-to-quality channel operates in the opposite direction: during acute uncertainty episodes, investors shift capital from risk assets into high-quality government and investment-grade bonds, compressing yields and supporting bond prices even as GEPU rises. Baele et al. (2020) quantified this mechanism, showing that during episodes where GEPU exceeds its 90th percentile, the stock–bond correlation turns negative and US Treasury returns average +0.8% per month — compared to +0.3% during normal periods — as capital flows from equities into safe-haven fixed income. Because GEPU drives up stock and commodity volatility but lowers bond risk through the flight-to-quality mechanism, Bitcoin’s relationships with these asset pairs are expected to have different signs (Bouri et al., 2021).

For Bitcoin specifically, the GEPU transmission mechanism carries an additional layer of complexity arising from the asset's rapid structural evolution over the estimation period. In the pre-institutional era prior to 2020, Bitcoin's investor base was predominantly retail and speculative, so its price dynamics were driven more by network sentiment and momentum than by macroeconomic fundamentals (Demir et al., 2018), and rising GEPU was associated with falling Bitcoin prices in a risk-off pattern. However, the subsequent entry of institutional investors, whose portfolio decisions are explicitly conditioned on macroeconomic signals, fundamentally changed this transmission. Wu et al. (2019) provided early evidence of the resulting instability through a time-varying parameter model, documenting that the Bitcoin–GEPU coefficient shifts sign across uncertainty regimes. Moreover, Bouri et al. (2022) added a frequency dimension, showing that the GEPU–Bitcoin relationship is negligible at daily frequencies but strengthens substantially over monthly and quarterly periods, directly validating the mixed-frequency MIDAS architecture and confirming that daily-frequency models cannot capture a transmission channel operating at a business-cycle horizon.

Taken together, the literature establishes GEPU as a systematic risk factor whose cross-asset effects are heterogeneous and structurally unstable, making it both valuable and difficult to use in hedging models. Its value lies in the possibility that GEPU-conditioned hedge ratios can improve variance reduction by allowing the long-run covariance between Bitcoin and conventional assets to respond to policy uncertainty (Fang et al., 2019). The difficulty is methodological, because GEPU is observed monthly while asset returns are measured daily, and aggregating returns to match the macro series would discard important short-run information (Baker et al., 2023). The Mixed-Data Sampling framework of Ghysels, Santa-Clara, and Valkanov (2004), later incorporated into GARCH-MIDAS and DCC-MIDAS by Engle et al. (2013) and Colacito et al. (2011), directly addresses this frequency mismatch by linking low-frequency macroeconomic uncertainty to high-frequency volatility and correlation dynamics without aggregation.

2.1.4 Mixed-frequency volatility models

2.1.4.1 The MIDAS framework

Resolving the frequency mismatch between daily returns and monthly uncertainty measures has become a central concern in the mixed-frequency financial literature. This literature progresses from the MIDAS regression of Ghysels et al. (2004) to GARCH-based extensions, providing a framework for linking high-frequency financial dynamics with low-frequency macroeconomic conditions.

At its root, the challenge in macro-finance forecasting is that key variables are observed at different frequencies, with asset returns available daily, policy uncertainty indices published monthly, and GDP released quarterly. The traditional remedy of aggregating high-frequency data to match the lowest-frequency variable is methodologically costly, since averaging daily returns to monthly observations discards the timing and sequential dynamics of the high-frequency series and introduces measurement error that biases coefficient estimates (Andreou, Ghysels & Kourtellos, 2010). The Mixed-Data Sampling (MIDAS) framework of Ghysels, Santa-Clara, and Valkanov (2004) solves this by allowing high- and low-frequency variables to enter the same regression directly, without aggregation.

The core MIDAS regression takes the form:

$$y_t = \beta_0 + \beta_1 \cdot \sum_{k=0}^{K-1} w(k;\theta)\, x_{t\cdot m-k}^{(m)} + \varepsilon_t$$

where y_t is a low-frequency daily dependent variable, $x_{t\cdot m-k}$ is the high-frequency regressor observed m times per low-frequency period, K is the number of high-frequency lags included, and *w(k; θ)* is a weighting function parameterized by a low-dimensional vector θ, normalized so that the weights sum to 1. The key advantage of MIDAS lies in its parsimony, since instead of estimating (K) separate lag coefficients, which would require many parameters and quickly create a curse-of-dimensionality problem, the model summarizes the full distributed-lag structure using only two or three parameters in θ, while still allowing the optimal lag-decay profile to be determined by the data (Ghysels et al., 2007).

Weighting Schemes

The choice of lag-weighting function has been a central point of refinement within the MIDAS literature, which has progressively favoured schemes that balance parsimony against the flexibility to represent diverse decay patterns (Ghysels, Sinko & Valkanov, 2007; Mogliani & Simoni, 2021). Two principal families of lag polynomial have become standard. The first is the Exponential Almon polynomial, proposed by Ghysels et al. (2004) by adapting the distributed-lag estimator of Almon (1965) to the nonlinear MIDAS context. It parameterizes each lag weight as:

$$w(k;\theta_1,\theta_2) = \frac{\exp(\theta_1 k + \theta_2 k^2)}{\sum_{j=0}^{K-1} \exp(\theta_1 j + \theta_2 j^2)}$$

where k = 0, 1, …, K−1 indexes the lags from most recent to most distant. The two parameters θ_1 and θ_2 jointly control the slope and curvature of the weight profile:

negative θ_2 produces a hump-shaped curve peaking at an intermediate lag, while $\theta_1 < 0$ with $\theta_2 = 0$ yields monotone decay from the most recent observation. The exponential form guarantees strictly positive weights for all parameter values, automatically satisfying the positivity constraint.

The second standard weighting scheme is the Beta polynomial, which parameterizes the weights through the Beta probability density function. Following Ghysels, Sinko, and Valkanov (2007), the Beta weights are defined as:

$$w(k;\alpha,\beta) = \frac{f\left(\frac{k}{K};\alpha,\beta\right)}{\sum_{j=0}^{K-1} f\left(\frac{j}{K};\alpha,\beta\right)}, \quad f(x;\alpha,\beta) = x^{(\alpha-1)}(1-x)^{(\beta-1)}$$

Each lag index k is rescaled to the unit interval via $x = k/K$, and $f(x;\alpha,\beta)$ is the Beta(α, β) density kernel evaluated at that point. Together, α and β determine the full geometry of the weight profile: $\alpha = \beta = 1$ yields uniform weights; $\alpha > 1$ and $\beta > 1$ produces a hump-shaped profile; $\alpha < 1$ with $\beta > 1$ gives monotone decay from short lags; and $\alpha > 1$ with $\beta < 1$ allows weights to rise toward longer lags. Relative to the Exponential Almon scheme, the Beta polynomial accommodates a richer variety of shapes, including U-shaped and flat-tailed profiles. In practice, the restricted Beta(α, 1) form is commonly imposed to reduce the parameter space while preserving monotone-decay flexibility.

In both cases, the weighting function is estimated jointly with the regression coefficients through non-linear least squares, allowing the lag-decay profile to be determined by the data rather than imposed in advance by the researcher (Ghysels et al., 2007). This matters for GEPU because policy uncertainty is not absorbed by markets immediately; its effects pass through investment, hiring, and portfolio-reallocation decisions that unfold gradually over time (Baker et al., 2016). Consistent with this logic, Bouri et al. (2020) showed that the GEPU–Bitcoin relationship becomes stronger at monthly and quarterly horizons, indicating that its effect is not purely contemporaneous. Consequently, this slow-moving transmission mechanism explains why MIDAS weighting is preferable to fixed-lag or aggregation-based approaches: it preserves high-frequency return information while allowing low-frequency uncertainty shocks to influence financial risk over time (Dai et al., 2022).

2.1.4.2 GARCH-MIDAS volatility decomposition

Engle, Ghysels, and Sohn (2013) combined the MIDAS weighting scheme with the GARCH framework to create the GARCH-MIDAS model. The key idea behind this model is that it allows daily return observations to be linked directly with macroeconomic variables sampled at lower frequencies, in order to examine how those macroeconomic drivers affect asset volatility. To do this, the model splits the total conditional variance into two parts that are multiplied together:

$$h_{i,t} = g_{i,t} \times \tau_{i,t}$$

Here, $g_{i,t}$ denotes the short-run component, which captures day-to-day volatility clustering, where large price movements tend to be followed by further large movements, through a standard GARCH(1,1) process. By contrast, $\tau_{i,t}$, represents the long-run component, which evolves slowly at monthly frequency and links persistent volatility to macroeconomic fundamentals through the MIDAS weighting scheme. Since the two components enter multiplicatively rather than additively, total conditional variance remains positive, while daily shocks are interpreted relative to the prevailing long-run volatility level.

This decomposition addresses a well-known feature of financial markets, documented since Engle and Lee (1999), that return volatility exhibits both short-lived bursts driven by recent shocks and longer-run swings driven by the business cycle and policy uncertainty, phenomena that operate at different frequencies and require different modeling approaches. By separating these two components, the GARCH-MIDAS model allows a precise question to be posed, namely whether a given macroeconomic variable such as GEPU carries information about the persistent component of asset volatility beyond what the short-run GARCH dynamics already capture. Engle et al. (2013) applied the model to US equity volatility using realized variance and industrial production as conditioning variables, finding that both carry significant information about long-run volatility and that the specification fits substantially better than standard GARCH. Conrad and Loch (2015) extended this to allow multiple low-frequency variables to enter the long-run equation simultaneously, the GARCH-MIDAS-X specification, showing that macroeconomic predictors carry incremental information beyond realized variance alone.

The model has since been validated across all the asset classes used in this study, confirming that the short-run/long-run split is a general feature of financial returns rather than something specific to US equities. For equities, Asgharian, Hou, and Javed (2013) found that macroeconomic variables explain roughly 15–25% of the long-run volatility component of international equity indices and reduce out-of-sample forecast errors by 8–12% relative to standard GARCH models. For commodities, Wang, Ma, and Wu (2015) applied the framework to crude oil and found that economic uncertainty variables improve long-run forecast accuracy by a similar margin. Most relevant to this study, Conrad, Custovic, and Ghysels (2018) applied the model to Bitcoin and found that, despite Bitcoin's extreme daily volatility (a daily standard deviation above 4%), about 30–40% of its total conditional variance comes from the slow-moving long-run component tied to global macroeconomic conditions. Bitcoin's volatility therefore contains a persistent, macro-sensitive part that a GARCH-MIDAS-X model can isolate and link to GEPU.

2.1.4.3 DCC-MIDAS-X dynamic correlation modeling

Colacito, Engle, and Ghysels (2011) extended the GARCH-MIDAS decomposition from univariate volatility to bivariate correlation structure, creating the DCC-MIDAS model. The key insight parallels the volatility decomposition: just as total conditional variance can be separated into a short-run GARCH component and a long-run MIDAS component, the conditional correlation between two assets can be separated into a short-run DCC component — following the standard Engle (2002) recursion among GARCH-MIDAS-standardized residuals — and a long-run correlation target that evolves at monthly frequency, driven by a MIDAS-weighted average of past realized correlations. This decomposition captures an empirically important feature of asset correlations that the standard DCC model misses: correlations between asset classes do not fluctuate randomly around a fixed unconditional mean but exhibit persistent, low-frequency swings that are driven by the business cycle and the macroeconomic policy environment (Güngör, 2021).

Building on this foundation, Conrad, Loch, and Rittler (2014) extended the DCC-MIDAS framework to its most general form — the DCC-MIDAS-X specification — by allowing low-frequency macroeconomic variables to enter the long-run correlation equation directly, alongside the MIDAS-weighted realized correlations. Applying it to the US stock–crude oil correlation over 1993–2011, they found that variables carrying information on current and future economic activity significantly predict the long-run stock-oil correlation, which moves counter-cyclically — turning negative during above-trend growth and rising before and during recessions. This established DCC-MIDAS-X as the natural framework for testing whether a macroeconomic variable such as GEPU shapes the long-run correlation between Bitcoin and conventional asset classes. Liu and Lee (2021) subsequently confirmed that macroeconomic uncertainty variables significantly influence the Bitcoin–gold long-run correlation and that the DCC-MIDAS-X model produces superior out-of-sample correlation forecasts relative to the standard DCC benchmark.

Subsequent work has confirmed that the sign and magnitude of the macro effect on long-run correlations are themselves asset-class-specific rather than uniform. Ding, Ji, Ma, and Zhai (2022) applied a DCC-MIDAS-X specification to high- and low-carbon assets and found that policy risk exerts a significantly negative effect on their long-run correlation, with the strength of the effect depending on the asset category — and showed that the resulting model produces hedging and optimal-weight portfolios that outperform standard benchmarks. In the cryptocurrency setting specifically, the most direct application of mixed-frequency uncertainty conditioning to Bitcoin to date remains Fang, Bouri, Gupta, and Roubaud (2019), who used GARCH-MIDAS and DCC-MIDAS models with the GEPU index to show that policy uncertainty carries significant information about Bitcoin's long-run volatility and its long-run correlations with global equities, commodities, and bonds, and that GEPU-conditioned hedge ratios improve hedging effectiveness relative to standard DCC benchmarks.

2.2 Hypothesis Development

Existing research on Bitcoin's hedging properties presents a contested and structurally incomplete picture. Early studies by Dyhrberg (2016) and Bouri et al. (2017) found meaningful diversification benefits across several asset classes, creating optimism about Bitcoin's portfolio role. Later evidence was more cautious. Smales (2019) argued that Bitcoin's extreme volatility makes it unsuitable for institutional hedging, while Conlon and McGee (2020) showed that Bitcoin fell sharply with equities during the COVID-19 crash of March 2020, failing as a safe haven when downside protection was most needed. This contrast suggests that Bitcoin's hedging capacity is regime-dependent rather than fixed. Consequently, models that ignore this regime dependence may therefore produce systematically misleading results.

Fang, Bouri, Gupta, and Roubaud (2019) represent the most directly relevant benchmark for addressing this problem. By applying the GARCH-MIDAS and DCC-MIDAS framework with the Baker et al. (2016) GEPU index as the conditioning variable, they provided the first formal evidence that policy uncertainty drives Bitcoin's long-run volatility and correlation dynamics, and that hedge ratios conditioned on GEPU outperform standard DCC benchmarks in terms of hedging effectiveness. Xiong, Shen, Yoon & Dong (2024) subsequently demonstrated that the GEPU transmission mechanism is not stable across the sample period, and that ignoring structural breaks in the MIDAS parameters generates materially biased estimates of the GEPU transmission effect and systematically distorts measured hedging effectiveness. Together, these two studies define the frontier of the existing literature — and its remaining gaps. Fang et al. (2019) did not account for structural instability in the GEPU–Bitcoin relationship, and their sample predates three of the four major uncertainty regimes examined here. On the other hand, Xiong et al. (2024) incorporated Bai–Perron structural breaks into a DCC-MIDAS-X model but applied it to the stock–exchange-rate correlation in China, leaving the cryptocurrency setting unexplored. Their design also provided no systematic decomposition of how much hedging improvement is attributable to GEPU conditioning versus structural-break accommodation, and offered limited out-of-sample or transaction-cost validation across equities, commodities, and bonds simultaneously.

Building on these gaps, the empirical analysis is organized around three core objectives:

(1) Establishing whether GEPU carries statistically significant information about the persistent long-run volatility component of Bitcoin and each conventional asset class, after controlling for realized variance;

(2) Assessing whether GEPU significantly shapes the long-run pairwise correlations between Bitcoin and each conventional asset class, and whether the direction of its effect differs across asset classes in a manner consistent with the transmission channels identified in the literature;

(3) Evaluating whether accounting for structural breaks in the GEPU transmission mechanism, and conditioning hedge ratios on both GEPU and the prevailing uncertainty regime, produces statistically and economically significant improvements in portfolio variance reduction relative to the standard DCC-MIDAS benchmark.

Four formal hypotheses are developed below, each corresponding directly to one of these objectives and generating a clear, falsifiable empirical prediction.

- *H1: The MIDAS slope coefficient on lagged GEPU (θ^{V}_{GEPU}) is statistically significant in the long-run volatility equation for Bitcoin, MSCI World, S&P GSCI, and PIMCO, after controlling for realized variance.*
- *H2: The MIDAS slope coefficient on lagged GEPU (θ^{C}_{GEPU}) is statistically significant in the long-run correlation equation for each Bitcoin–conventional asset pair, with the direction of the effect differing across asset classes in a manner consistent with the asset-class-specific transmission channels documented in the literature.*
- *H3: Allowing the MIDAS slope coefficients to shift across Bai–Perron structural break regimes significantly improves model fit relative to the single-regime specification, as indicated by Wald joint tests rejecting the null hypothesis of zero break dummy coefficients for both the long-run volatility and long-run correlation equations.*

- *H4: The hedging effectiveness of the full GEPU-augmented DCC-MIDAS-X model with structural breaks is statistically significantly higher than that of the benchmark DCC-MIDAS model for each Bitcoin–conventional asset pair, both in-sample and out-of-sample, and after adjusting for transaction costs at representative levels of 10, 25, and 50 basis points.*

The first hypothesis (H1) draws on the GARCH-MIDAS-X literature, which shows that low-frequency macroeconomic variables explain the persistent component of asset volatility beyond short-run GARCH effects (Engle, Ghysels, and Sohn, 2013; Conrad and Loch, 2015). In the cryptocurrency context, Fang et al. (2019) provide direct evidence that GEPU matters for Bitcoin, equities, and commodities. However, their sample ends in early 2018 and excludes the trade war, COVID-19, the tightening cycle, and the post-ETF period. As Bitcoin has become more institutionally integrated, the effect of GEPU may differ across assets and regimes. H1 therefore tests whether GEPU remains a significant driver of long-run volatility across Bitcoin, equities, commodities, and bonds over the extended 2016–2025 sample.

The second hypothesis (H2) addresses the DCC-MIDAS-X stage and tests whether GEPU affects Bitcoin's long-run correlations with conventional assets in an asset-class-specific way. Long-run correlations are not fixed but evolve with macroeconomic conditions (Colacito, Engle, and Ghysels, 2011; Conrad, Loch, and Rittler, 2014). For Bitcoin, Fang et al. (2019) — using a pre-2018 sample — report that GEPU weakens the Bitcoin–bond correlation while strengthening its equity and commodity correlations. However, Bitcoin's post-2020 institutionalization may have altered this pattern: as macro-conditioned institutional capital replaced retail flows, the equity-pair sign could plausibly reverse. H2 therefore does not impose Fang's signs a priori; it tests only (i) whether θ^{C}_{GEPU} is statistically significant for each pair and (ii) whether the sign differs across asset classes, treating any reversal of the equity-pair sign relative to Fang et al. (2019) as an empirical finding about Bitcoin's changing market structure rather than a contradiction.

The third hypothesis (H3) is motivated by the structural break literature and the macroeconomic record of the sample window. Bai and Perron (2003) provide the standard procedure for detecting multiple breaks without imposing dates exogenously, and Ang and Timmermann (2012) show that parameter stability across market cycles is the exception rather than the rule. For MIDAS models of Bitcoin specifically, Fang et al. (2019) document parameter instability, while Xiong et al. (2024) show that ignoring breaks biases MIDAS slope estimates and distorts hedging effectiveness. Because the sample spans four distinct regimes — the US–China trade war, the COVID-19 shock, the 2022–2023 tightening cycle, and the post-ETF phase — discrete shifts in GEPU transmission are likely. H3 therefore tests whether incorporating these shifts through the Bai–Perron framework significantly improves model fit in both the volatility and correlation equations.

The fourth hypothesis (H4) translates the volatility and correlation findings from H1–H3 into actual portfolio performance. If GEPU contains useful information about the long-run covariance between Bitcoin and conventional assets, then hedge ratios that incorporate this information should reduce hedged portfolio variance more effectively than benchmark models that ignore it. Consistently, Fang et al. (2019) show that GEPU-conditioned DCC-MIDAS hedge ratios improve Bitcoin hedging, although the gains are modest, while Xiong et al. (2024) suggest that incorporating structural breaks can further reduce parameter bias and improve hedging measurement. Therefore, H4 tests whether the full regime-aware DCC-MIDAS-X model delivers stronger hedging effectiveness, both in-sample and out-of-sample, after accounting for transaction costs of 10, 25, and 50 basis points.

Overall, the four hypotheses convert the literature gap into a coherent empirical testing strategy. H1 and H2 examine whether GEPU explains Bitcoin's long-run volatility and correlations with conventional assets. H3 tests whether this transmission changes across structural regimes, while H4 evaluates whether these modelling gains translate into real portfolio risk reduction. Together, the hypotheses move the analysis from macro-financial transmission to hedging performance, allowing the thesis to assess whether a regime-aware DCC-MIDAS-X framework improves Bitcoin hedging in both statistical and economic terms.

Summary of Chapter 2

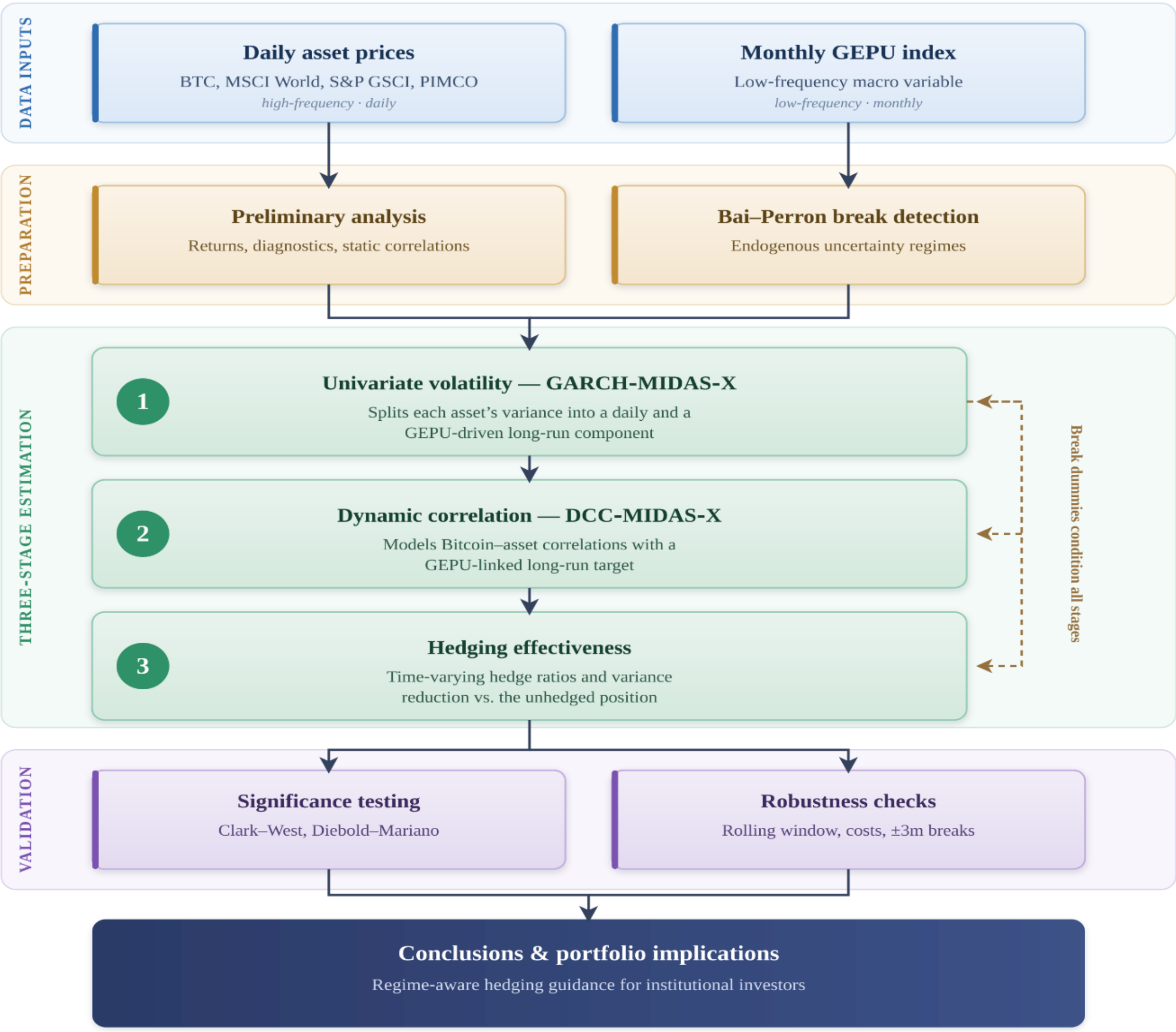
Bitcoin was originally expected to stand apart from the forces that move conventional financial markets, since its decentralized structure appeared to make it less dependent on central banks, government policy, and macroeconomic cycles; however, the literature increasingly shows that its relationship with equities, commodities, and bonds is far more unstable than this early view suggested, with its hedging value changing across both asset classes and market regimes. Accordingly, global economic policy uncertainty becomes central to this instability because it affects long-run volatility and cross-asset correlations through slow-moving macro-financial channels, meaning that Bitcoin's portfolio role cannot be assessed only through short-run price movements or average correlations. In this context, Fang et al. (2019) provide the key starting point by showing that GEPU contains meaningful information about Bitcoin's long-run volatility, correlations, and hedging effectiveness within a mixed-frequency GARCH-MIDAS and DCC-MIDAS framework, although their model treats this transmission channel as constant over time. Crucially, this assumption becomes increasingly restrictive in the post-2018 period, when the trade war, COVID-19 crisis, monetary tightening cycle, and growing institutional participation in Bitcoin markets created distinct uncertainty regimes with potentially different GEPU effects. More recently, Xiong et al. (2024) address this weakness by demonstrating that ignoring structural breaks in MIDAS parameters can distort coefficient estimates and understate hedging performance, which means that the GEPU channel identified by Fang et al. (2019) cannot be measured reliably unless its regime instability is explicitly modeled. Therefore, building on this gap, the thesis combines GARCH-MIDAS-X and DCC-MIDAS-X models with Bai–Perron structural breaks across Bitcoin–equity, Bitcoin–commodity, and Bitcoin–bond pairs, with the purpose of testing whether regime-aware GEPU conditioning improves the modelling of Bitcoin's volatility and correlations and whether this improvement translates into stronger portfolio risk reduction over the 2016–2025 sample.

CHAPTER 3: METHODOLOGY AND DATA COLLECTION

3.1 Methodology

3.1.1 Research design

Figure 3.1: Research design



(Sources: Xiong et al. (2024), author's compilation)

To evaluate whether conditioning hedge ratios on global economic policy uncertainty (GEPU) reduces portfolio variance, this study employs a well-established empirical design from the mixed-frequency volatility and hedging literature (Fang et al., 2019; Xiong et al., 2024). The framework follows the GARCH-MIDAS and DCC-MIDAS tradition of Engle et al. (2013) and Colacito et al. (2011), which separates volatility and correlation into a fast-moving daily component and a slow-moving monthly component linked to macroeconomic conditions. It is further extended with the Bai and Perron (2003) structural-break procedure, allowing the effect of policy uncertainty to shift across distinct uncertainty regimes rather than remaining constant over the full sample.

The research process advances from two data streams — daily asset prices and the monthly GEPU index — through three sequential estimation stages: univariate volatility modeling, dynamic correlation modeling, and hedging effectiveness evaluation. The results are then subjected to significance testing and robustness checks before the analysis concludes with portfolio implications for designing a regime-specific Bitcoin hedging strategy.

3.1.2 Univariate Volatility: GARCH–MIDAS–X model with structural breaks

Before examining, through the DCC-MIDAS model, the potential role of GEPU in shaping the dynamic correlations between Bitcoin and conventional asset classes, the author first employs the univariate GARCH-MIDAS framework of Engle et al. (2013), as further developed by Conrad and Loch (2015), to directly investigate how the short- and long-run volatility dynamics of Bitcoin, MSCI, S&P GSCI, and the PIMCO index are affected by the GEPU. In this approach, the short-run volatility component is captured by a mean-reverting daily GARCH process, while the long-run volatility component is modelled as a function of low-frequency macro-financial variables, namely realized volatility (RV) and GEPU, through a MIDAS beta weighting scheme. The GARCH-MIDAS model is specified as follows:

$$r_{i,t} = \mu_i + \varepsilon_{i,t} = \mu_i + \sqrt{\tau_t g_{i,t}}\, \xi_{i,t} \tag{1}$$

$$\xi_{i,t} | \Omega_{i,t-1} \sim N(0,1)$$

Here, $\Omega_{i,t-1}$ denotes the information set available up to time t - 1, where $r_{i,t}$ denotes the log returns of asset i in a period t = 1,…, T; μ_i is the unconditional mean of return; $\xi_{i,t}$ is the standardized residual or innovation; $g_{i,t}$ is the short-run variance component; and τ_t is defined as the long-run component evolving at monthly frequency[3]. Accordingly, GARCH-MIDAS decomposes asset's total variance as the

[3] Each univariate GARCH-MIDAS-X model is estimated separately for each asset. For notational brevity, the asset subscript i is suppressed on τ_t and on all associated parameters (m, θ_{RV}, θ_X, ω); each is understood to be asset-specific throughout Section 3.1

product of a transitory daily component and a persistent monthly component, as in Engle et al. (2013): $\sigma^2 = \tau_t \cdot g_{i,t}$.

The conditional variance dynamics of the component $g_{i,t}$ accounting for daily fluctuations assumed to be short-lived (Chordia et al., 2002) follows a standard GARCH (1,1) process:

$$g_{i,t} = (1 - \alpha_i - \beta_i) + \alpha_i \frac{(r_{i,t-1} - \mu_i)^2}{\tau_t} + \beta_i g_{i,t-1} \tag{2}$$

with the constraints of $\alpha_i > 0$, $\beta_i \geq 0$ and $\alpha_i + \beta_i < 1$.

Long-run volatility: Benchmark specification

The author then parameterizes the persistent volatility component by linking long-run variance to lagged realized volatility, with the temporal contribution of past monthly shocks disciplined through a MIDAS polynomial:

$$log(\tau_t) = m_v + \theta_{RV} \sum_{k=1}^{K} \varphi_k (\omega_v) RV_{t-k} \tag{3}$$

$$RV_t = \sum_{d=1}^{N_t} r_{d,t}^2 \tag{4}$$

where *m* is the intercept term, measuring unconditional long-term volatility and K is the number of lags over which volatility is smoothed[4]. Regarding Equation (4), RV_t is the monthly realized variance, computed as the sum of squared daily log returns within calendar month t, N_t is the number of trading days in that month.

Long-run volatility: Extended specification with GEPU

To investigate the impact of global economic policy uncertainty (GEPU) on the long-run variance, the author extends Equation (3) by incorporating GEPU as an additional low-frequency variable alongside RV (Asgharian et al., 2013):

[4] The optimal lag length K = 12 months is selected based on the AIC and likelihood ratio test results reported in Table 4.4, consistent with Fang et al. (2019) and Xiong et al. (2024).

$$log(\tau_t) = m_v + \theta_{i,RV} \sum_{k=1}^{K} \varphi_k(\omega_v^{RV})\, RV_{t-k} + \theta_{i,X} \sum_{k=1}^{K} \varphi_k(\omega_v^{X})\, X_{t-k}^{GEPU} \quad (5)$$

where X_{t-k}^{GEPU} represents the lagged GEPU index, and $\theta_{i,X}$ captures its influence on the long-run volatility of asset *i*. A positive and statistically significant $\theta_{i,X}$ indicates that higher global policy uncertainty is associated with higher persistent volatility, after controlling for realized volatility.

The MIDAS beta weighting scheme in specification (3) is defined by the beta lag polynomial (Zhou et al., 2019, Liu and Lee, 2021):

$$\varphi_k(\omega_v^i) = \frac{(1 - \frac{k}{K})^{\omega_v^s - 1}}{\sum_{s=1}^{K}(1 - \frac{s}{K})^{\omega_v^s - 1}}, \forall k = 1, \ldots, K \quad (6)$$

The MIDAS beta weights, $\varphi_k(\omega_1, \omega_2)$, determine the relative importance assigned to each lagged monthly observation in the long-run component. Following Dong et al. (2023) and Xiong et al. (2024), the author imposes $\omega_1 = 1$, so that ω_2 controls the shape and decay rate of the weighting scheme. Larger values of ω_2 place greater weight on more recent observations, while smaller values imply a smoother and more persistent lag distribution over time.

Structural breaks in the long-run volatility component

Policy uncertainty may affect asset volatility differently across crisis and non-crisis regimes. To account for this asymmetry, the long-term volatility component is further modified by introducing dummy variables that capture regime shifts in GEPU series, identified through sequential structural break testing (Bai and Perron, 2003):

$$\begin{aligned} log(\tau_t) = (m_v + \sum_j m_{v,j} D_j) + \left(\theta_{RV} + \sum_j \theta_{RV,j} D_j\right) \sum_{k=1}^{K} \varphi_k\,(\omega_v^{RV}) RV_{t-k} \\ + \left(\theta_X + \sum_j \theta_{X,j}\, D_j\right) \sum_{k=1}^{K} \varphi_k\,(\omega_v^{X})\, X_{t-k}^{GEPU} \end{aligned} \quad (7)$$

Following Xiong et al. (2024), the author applies the sequential sup-F procedure of Bai and Perron (2003) to detect structural changes in the GEPU series, dividing the sample into J + 1 subsample. The first subsample period, preceding the first structural break, is designated as the base subsample regime (subsample 0) with $j = 0$ to avoid the dummy variable trap. Here, j represents the subsample period, meaning that D_1 is equal to one between the first and second breakpoints (called subsample 1) and zero otherwise. The settings for D_2 to D_n follow the same convention. The coefficients $\theta_{RV,j}$ and $\theta_{X,j}$ represent the incremental effects on the parameter θ_{RV} and θ_X during subsample period j, relative to the base regime.

Break detection, volatility and correlation modelling are conducted in separate steps, as is standard in the DCC-MIDAS literature with structural breaks (Conrad & Loch, 2015; Xiong et al., 2024). Joint estimation of break locations alongside the full GARCH-MIDAS-X parameter vector (up to 15–20 parameters per asset pair when structural-break dummies are included) would render the optimization non-smooth and substantially increase the dimensionality of the search space. A known limitation is that break-point estimation uncertainty is not propagated into the second-stage inference on θ_{RV} and θ_X (Elliott and Müller, 2007), potentially understating standard errors by 5–15% in finite samples of comparable size. The author mitigates this via (i) robust (sandwich) standard errors of Bollerslev and Wooldridge (1992) and (ii) a sensitivity check in which each break date is perturbed by ±3 months (approximately ±60 trading days) to verify that the sign, magnitude, and significance of the MIDAS slope coefficients remain stable across alternative regime datings.

Distributional assumption and diagnostic checking

The conditional normality assumption imposed on the standardized innovations $\xi_{i,t}$ in Equation (1) is adopted for estimation convenience rather than as a literal description of the data-generating process. Cryptocurrency and commodity returns, in particular, are well documented to exhibit excess kurtosis and, in some cases, skewness (Katsiampa, 2017; Baur et al., 2018), which violates the Gaussian assumption. Nevertheless, the quasi-maximum likelihood (QML) estimator remains consistent and asymptotically normal for the GARCH parameters even when the true innovation distribution is non-Gaussian, provided the conditional mean and variance are correctly specified (Bollerslev and Wooldridge, 1992). Accordingly, all reported standard errors are based on the robust (sandwich) covariance matrix of Bollerslev and Wooldridge (1992), which accounts for potential misspecification of the conditional density. This approach is standard in the GARCH-MIDAS literature (Engle et al., 2013; Conrad and Loch, 2015) and ensures that inference on the MIDAS slope parameters θ_{RV} and θ_X is reliable regardless of the precise distributional form of the innovations. To assess model adequacy, the author applies the Ljung-Box Q-statistic to the standardized residuals and their squares at lags 10 and 20, testing for remaining serial correlation and ARCH effects and the Engle (1982) LM test further verifies the absence of residual heteroskedasticity.

3.1.3 Dynamic Conditional Correlation: DCC–MIDAS–X model with structural breaks

Before modelling the dynamic dependence, the author defines the bivariate return system underlying the DCC-MIDAS-X framework. Let the bivariate return vector at time t be denoted by $r_t = (r_{i,t}, r_{j,t})'$, where $r_{i,t}$ and $r_{j,t}$ represents the return on Bitcoin and the conventional asset indices, respectively[5]. Let $\Omega_{t\text{-}1}$ denote the information set available up to time t−1. Conditional on this information set, the return vector is assumed to have conditional mean $\mu_t = E(r_t \mid \Omega_{t\text{-}1})$, and the innovation vector filtered in Equation (1) is defined as $\varepsilon_t = r_t - \mu_t$, with $E(\boldsymbol{\varepsilon}_t \mid \Omega_{t\text{-}1}) = 0$.The conditional covariance matrix of these innovations is denoted by $H_t = Var(\varepsilon_t \mid \Omega_{t\text{-}1})$. Following the DCC framework of Engle (2002), this conditional covariance matrix can then be decomposed as $H_t = D_t \cdot R_t \cdot D_t$, where D_t is a diagonal matrix containing the conditional standard deviations of returns on the diagonal obtained from the first-stage GARCH-MIDAS-X models, and R_t is the dynamic conditional correlation matrix of the standardized return residuals of the univariate return.

$$R_t = \begin{pmatrix} 1 & \rho_{ij,t} \\ \rho_{ij,t} & 1 \end{pmatrix} \quad \text{and} \quad D_t = \begin{pmatrix} h_{i,t}^{1/2} & 0 \\ 0 & h_{j,t}^{1/2} \end{pmatrix}.$$

[5] Hereafter, i denotes Bitcoin, while j denotes one of the conventional asset indices considered in the analysis, namely MSCI for global equities, PIMCO for bonds, and S&P GSCI for commodities (each Bitcoin–index pair is estimated separately).

Building on this structure, the DCC-MIDAS model of Colacito et al. (2011) extends GARCH-MIDAS to dynamic correlations by decomposing them into short- and long-run components. Following Conrad et al. (2014), this study allows the long-run correlation to depend on lagged GEPU, making the framework suitable for mixed-frequency data, where asset returns are daily but GEPU is monthly. Compared with alternative multivariate GARCH models such as BEKK and CCC-GARCH, DCC-MIDAS is more appropriate because it explicitly links long-run correlations to low-frequency macro-financial variables. Structural breaks are further incorporated to allow the GEPU–correlation relationship to vary across uncertainty regimes.

Benchmark DCC-MIDAS model

In the bivariate Bitcoin–asset system, the dynamic conditional correlation is defined as $R_t = diag(Q_t)^{-1/2} * Q_t * diag(Q_t)^{-1/2}$, where Q_t is a short-term quasi-correlation matrix. Its off-diagonal element, $q_{ij,t}$, follows a GARCH(1,1)-type recursion that mean-reverts to the slowly moving long-run correlation component, $\overline{\rho}_{ij,\tau}$:

$$q_{ij,t} = (1 - a - b)\overline{\rho}_{ij,\tau} + a\xi_{i,t-1}\xi_{j,t-1} + bq_{ij,t-1} \tag{8}$$

Here, $\xi_{i,t-1}$ and $\xi_{j,t-1}$ are the standardized residuals from the first-stage volatility models (GARCH-MIDAS). The parameters a and b capture, respectively, the short-run response to new correlation shocks and the persistence of past correlation dynamics, subject to a ≥ 0, b ≥ 0, and a + b < 1. The monthly component $\overline{\rho}_{ij,\tau}$ acts as the slowly moving long-run correlation target in the Colacito et al. (2011) DCC-MIDAS framework and is specified as:

$$\overline{\rho}_{ij,\tau} = \sum_{\mathrm{k}=1}^{\mathrm{K}_{\mathrm{c}}^{\mathrm{ij}}} \varphi_k(\omega_r^{ij})\, c_{ij,t-k} \tag{9}$$

where $\varphi_k(\omega_r^{ij})$ represents the weight kernel function, mathematically formalized in Equation (6), and c_{ij} denotes the monthly realized correlation between assets i and j, expressed as:

$$c_{ij,t} = \frac{\sum_{k=t-N_c^{ij}}^{t} \xi_{i,k}\xi_{j,k}}{\sqrt{\sum_{k=t-N_c^{ij}}^{t} \xi_{i,k}^2}\sqrt{\sum_{k=t-N_C^{ij}}^{t} \xi_{j,k}^2}} \tag{10}$$

Equation (10) computes realized correlation from standardized residuals within the relevant historical window, where N_c^{ij} denotes the number of trading days used in the monthly correlation measure. If $\overline{\rho}_{ij,\tau}$ is constant, the DCC-MIDAS specification collapses to the standard DCC model.

For the bivariate case, the long-run correlation target enters the following matrix:

$$S_t = \begin{pmatrix} \overline{\rho}_{ii,t} & \overline{\rho}_{ij,t} \\ \overline{\rho}_{ij,t} & \overline{\rho}_{jj,t} \end{pmatrix} = \begin{pmatrix} 1 & \overline{\rho}_{ij,t} \\ \overline{\rho}_{ij,t} & 1 \end{pmatrix}$$

The DCC correlation-driving process Q_t is correspondingly represented as:

$$Q_t = \begin{pmatrix} q_{ii,t} & q_{ij,t} \\ q_{ij,t} & q_{jj,t} \end{pmatrix},$$

Equations (8)–(10) show that Q_t is updated through three elements: the long-run correlation target, the most recent standardized-residual shock, and the previous quasi-correlation matrix. Under the restrictions $a \geq 0$, $b \geq 0$, and $a + b < 1$, this recursion remains stable and mean-reverting. The final normalization step then transforms Q_t into a valid conditional correlation matrix with unit diagonal elements.

GEPU-augmented DCC-MIDAS-X model

Following Conrad et al. (2014), the DCC-MIDAS-X model allows low-frequency macro-finance variables to enter the long-run correlation component. In this study, GEPU is introduced as the main covariate, while structural-break dummies allow its effect to vary across uncertainty regimes.

The key object is the long-run correlation target $\overline{\rho}_{ij,\tau}$, which is made time-varying through MIDAS-weighted realized correlations and macroeconomic covariates.

Because a linear covariate specification may generate fitted correlations outside the admissible interval [−1, 1], the author follows Christodoulakis and Satchell (2002) and models the Fisher-z transformed long-run correlation. Mapping the transformed value back through the inverse Fisher transformation ensures that $\overline{\rho}_{ij,\tau}$ remains a valid correlation coefficient:

$$\overline{\rho}_{ij,\tau} = \frac{\exp\left(2\overline{\mathrm{z}}_{\mathrm{ij},\tau}\right) - 1}{\exp\left(2\overline{\mathrm{z}}_{\mathrm{ij},\tau}\right) + 1} \tag{11}$$

In the benchmark specification, the transformed long-run correlation is driven only by MIDAS-weighted realized correlations:

$$\overline{z}_{ij,\tau} = m_c + \theta_{RC} \sum_{k=1}^{K_c} \varphi_k(\omega_C^{RC})\, RC_{t-k} \tag{12}$$

The GEPU-augmented specification then adds lagged GEPU as a macro-finance covariate in the long-run correlation equation, following Conrad et al. (2014):

$$z_{ij,\tau} = m_c + \theta_{RC} \sum_{k=1}^{K_c} \varphi_k\, (\omega_C^{RC})\, RC_{t-k} \; + \; \theta_X \sum_{k=1}^{K_c} \varphi_k\, (\omega_c^{X}) X_{t-k} \tag{13}$$

The coefficient θ_X measures the marginal effect of GEPU on the Fisher-z transformed long-run correlation between Bitcoin and the conventional asset. A positive θ_X indicates that higher policy uncertainty strengthens the long-run correlation, whereas a negative θ_X implies a weakening of long-run dependence.

Structural breaks in the long-run correlation component

Then, to capture regime dependence, the extended specification interacts the intercept and MIDAS slope coefficients with structural-break dummies dated from the GEPU series:

$$z_{ij,\tau} = (m_c + \sum_j m_{c,j} D_j) + \left(\theta_{RC} + \sum_j \theta_{RC,j} D_j\right) \sum_{k=1}^{K_c} \varphi_k\,(\omega_C^{RC}) RC_{t-k} \\ + \left(\theta_X + \sum_j \theta_{X,j} D_j\right) \sum_{k=1}^{K_c} \varphi_k\,(\omega_c^{X}) X_{t-k}^{GEPU} \tag{14}$$

Finally, the time-varying conditional correlation between the standardized residuals can be recovered from the estimated DCC-MIDAS-X model as follows:

$$\rho_{ij,t} = \frac{q_{ij,t}}{\sqrt{q_{ii,t} q_{jj,t}}} \tag{15}$$

In Equation (15), $q_{ij,t}$, $q_{ii,t}$ and $q_{jj,t}$ are elements of the correlation-driving matrix Q_t. This matrix is not a conditional covariance matrix; it is an auxiliary process that governs raw correlation dynamics. The normalization in Equation (15) rescales Q_t so that the resulting R_t has unit diagonal elements and admissible off-diagonal correlations.

The conditional covariance matrix is subsequently recovered by combining the estimated dynamic conditional correlations with the conditional variances obtained from the first-stage GARCH-MIDAS-X models: $H_t = D_t \cdot R_t \cdot D_t$, where $D_t = diag\left(\sqrt{h_{1,t}}, \sqrt{h_{2,t}}, \dots, \sqrt{h_{N,t}}\right)$, and $h_{1,t} = \tau_t \cdot g_{i,t}$ denotes the conditional variance of asset i. Therefore, the conditional covariance between Bitcoin and asset j is given by: $h_{Bj,t} = \rho_{Bj,t}\sqrt{h_{B,t} h_{j,t}}$. The auxiliary matrix Q_t governs latent correlation dynamics, R_t converts these dynamics into valid conditional correlations, and H_t combines those correlations with asset-specific volatilities to obtain economically meaningful covariances. Accordingly, R_t is used to analyse Bitcoin's time-varying dependence with conventional assets, while H_t provides the variance–covariance inputs required for optimal hedge-ratio estimation.

3.1.4 Bitcoin's hedging effectiveness

In this section, the author evaluates whether incorporating GEPU improves the hedging performance of Bitcoin against three conventional asset indices (global equities, bonds, and commodities). The conventional asset index is treated as the long position to be hedged, while Bitcoin is treated as the hedging instrument. This structure follows the hedge-testing logic of Baur and Lucey (2010) and Bouri et al. (2017), and is consistent with the research question of whether Bitcoin can reduce the risk of traditional asset exposures.

The return on a hedged portfolio is denoted by:

$$R_{H,t} = R_{I,t} - \gamma_t R_{B,t} \tag{16}$$

where $R_{I,t}$ and $R_{B,t}$ are the return on three indices and Bitcoin position at time t, respectively; and γ_t is the hedge ratio. Conditional on $\Omega_{i,\,t-1}$, the variance of the hedged portfolio is:

$$Var\left(R_{H,t} \mid \Omega_{t-1}\right) = Var\left(R_{I,t} \mid \Omega_{t-1}\right) - 2\gamma_t Cov\left(R_{I,t}, R_{B,t} \mid \Omega_{t-1}\right) + \gamma_t^2 Var\left(R_{B,t} \mid \Omega_{t-1}\right)$$

(17)

The optimal hedge ratios (OHRs) are defined as the value of γ_t , which minimizes the conditional variance of the hedged portfolio, as follows:

$$\gamma_t^* \mid \Omega_{t-1} = \frac{Cov(R_{I,t}, R_{B,t} \mid \Omega_{t-1})}{Var(R_{B,t} \mid \Omega_{t-1})} \tag{18}$$

The conditional variances and covariance are obtained from 2x2 the symmetric conditional covariance matrix H_t, estimated by the DCC-MIDAS-X model. The diagonal elements of H_t provide $Var\left(R_{I,t} \mid \Omega_{t-1}\right)$ and $Var\left(R_{B,t} \mid \Omega_{t-1}\right)$, while the off-diagonal element provides $Cov\left(R_{I,t}, R_{B,t} \mid \Omega_{t-1}\right)$.

In comparing the hedging performances, the author uses the hedging effectiveness (HE) index. An effective hedging strategy should have a noticeable impact on the risk (standard deviation) without creating a noticeable impact on the return (mean) when included in a portfolio. The HE index is defined as:

$$HE = \frac{Var_{unhedged} - Var_{hedged}}{Var_{unhedged}} \tag{19}$$

where $Var_{unhedged} = Var(R_{I,t})$ is the variance of the unhedged index position; and $Var_{hedged} = Var(R_{H,t})$ is the variance of the hedged portfolio. A higher HE ratio indicates a superior risk-reduction strategy.

Equations (16)–(19) are applied separately to each Bitcoin–index pair, yielding three sets of time-varying hedge ratios and hedging effectiveness measures. For each pair, the author compares four nested specifications spanning both the realized-correlation baseline and its GEPU-augmented extension, each estimated without and with Bai–Perron structural breaks: *(i)* a baseline DCC-MIDAS model driven only by lagged realized correlations (Eq. 12), and *(ii)* the extended DCC-MIDAS-X model with structural breaks incorporating GEPU (Eq. 14). The incremental risk reduction attributable to GEPU is quantified as $\Delta HE = HE_X - HE_{base.}$

To evaluate the statistical significance of the hedging improvement, the author defines the loss differential as $d_t = (R^{base}_{H,t})^2 - (R^{X}_{H,t})^2$, where the squared hedged return serves as a proxy for the conditional portfolio variance on day t. Since the baseline DCC-MIDAS and the GEPU-augmented DCC-MIDAS-X specifications are nested — the former obtains under the restriction $\theta_X = 0$ — the standard Diebold and Mariano (1995) statistic is known to favour the smaller model in finite samples due to the noise introduced by estimating the additional parameters of the nesting model (Clark and McCracken, 2001). The author therefore adopts the adjusted mean-squared-prediction-error (MSPE-adj) statistic of Clark and West (2007) as the primary test, which corrects for this finite-sample bias and yields approximately standard-normal inference under the null of equal predictive accuracy. The null hypothesis $H^0: E[d_t] \leq 0$ is tested against the one-sided alternative $H_1: E[d_t] > 0$, where rejection indicates that conditioning the hedge ratio on GEPU produces a statistically significant reduction in portfolio variance. As a supplement, the author also reports the unadjusted Diebold and Mariano (1995) statistic with Newey–West heteroskedasticity- and autocorrelation-consistent standard errors, noting that its size may be mildly conservative in the nested setting.

Out-of-sample evaluation

To guard against the possibility that the in-sample hedging improvement is driven by overfitting, the author complements the full-sample analysis with an out-of-sample exercise. Specifically, the author adopts a rolling-window scheme in which the first T_0 trading days serve as the initial estimation sample and the remaining $P = T - T_0$ days constitute the evaluation period. The initial estimation window is set to $T_0 = \lfloor 0.70 \times T \rfloor$, reserving approximately 30% of the sample for out-of-sample evaluation. And the robustness is constructed to alternative splits (60/40 and 80/20). At each date $t \in \{T_0 + 1, \ldots, T\}$, the DCC-MIDAS and DCC-MIDAS-X models are re-estimated using the most recent T_0 observations, and the resulting one-step-ahead conditional covariance forecasts are used to construct the optimal hedge ratio γ_t^* via Equation (18). The hedged portfolio return $R_{H,t}$ (Eq. 16) is then computed using this ex-ante hedge ratio and the realised returns on date t. The out-of-sample hedging effectiveness is calculated from the sequence of realised hedged returns over the evaluation period, and the Diebold–Mariano and Clark–West tests described above are applied to the out-of-sample loss differentials.

Transaction costs

Because the optimal hedge ratio γ_t^* changes daily, rebalancing the hedged portfolio incurs trading costs. To assess whether the GEPU-augmented model retains its advantage after accounting for these frictions, the author adopts a proportional transaction cost framework. The net hedged portfolio return under transaction costs is defined as: $R_{H,t}^{net} = R_{I,t} - \gamma_t^* R_{B,t} - c \mid \gamma_t^* - \gamma_{t-1}^* \mid$ where c denotes the one-way proportional transaction cost and $|\gamma_t^* - \gamma_{t-1}^*|$ captures the turnover from rebalancing the Bitcoin hedge position. Daily rebalancing provides a conservative cost assumption because it maximizes hedge turnover, although less frequent rebalancing may reduce trading costs at the expense of lower hedge precision. The hedging effectiveness measures (Eq. 19), and the forecast comparison tests are then recomputed using $R_{H,t}^{net}$. Following the cryptocurrency trading-cost literature (Platanakis and Urquhart, 2019), this study considers transaction cost levels of $c \in \{10, 25, 50\}$ basis points.

3.1.5 Estimation method

The parameters of the DCC-MIDAS-X model are estimated by quasi-maximum likelihood (QML) method using the two-step procedure of Engle (2002), adapted to the MIDAS setting by Colacito et al. (2011). The composite log-quasi-likelihood decomposes as $\mathcal{L} = \mathcal{L}_V + \mathcal{L}_C$, where:

$$\mathcal{L}_V = -{}^{1}/_{2}\sum_{t=1}^{T}(n\log(2\pi) + \log|D_t^2| + \varepsilon_t' D_t^{-2}\varepsilon_t) \tag{20a}$$

$$\mathcal{L}_C = -{}^{1}/_{2}\sum_{t=1}^{T}(\log|R_t| + \xi_t' R_t^{-1}\xi_t - \xi_t'\xi_t) \tag{20b}$$

with n=2 in the bivariate case, $D_t = diag\left(\sqrt{\mathrm{h}_{1,t}}, \sqrt{\mathrm{h}_{2,t}}\right)$, $\varepsilon_t = r_t - \mu_t$, and ξ_t the vector of standardised residuals from the first stage.

Stage 1 (Volatility): For each asset separately, the GARCH-MIDAS-X parameters $\Theta^V = \{\alpha, \beta, m_V, \theta_{RV}, \theta_X^V, \omega^{RV}, \omega^{X,V}\}$ and associated structural-break coefficients are estimated by maximising $\mathcal{L}_V$. The estimated conditional variances $\hat{h}_{i,t} = \hat{\tau}_t \cdot \hat{g}_{i,t}$ then yield the standardised residuals $\hat{\xi}_{i,t} = D_t^{-1}\varepsilon_t = \frac{r_{i,t} - \hat{\mu}_i}{\sqrt{\hat{\tau}_t \cdot \hat{g}_{i,t}}}$.

Stage 2 (Correlation): Treating $\hat{\xi}_{i,t}$ as data, the DCC-MIDAS-X parameters $\Theta^C = \{a, b, m_c, \theta_{RC}, \theta_X^C, \omega^{RC}, \omega^{X,C}\}$ and associated structural-break coefficients are estimated by maximising $\mathcal{L}_C$.

Two remarks are in order. First, the two-step QML estimator is consistent under standard regularity conditions (Engle, 2002; Colacito et al., 2011). All reported standard errors are based on the robust sandwich covariance matrix of Bollerslev and Wooldridge (1992), which accounts for both potential misspecification of the conditional density and first-stage estimation uncertainty. Second, the lag lengths of the MIDAS filters (K for the volatility component and K_c for the correlation component) are selected by likelihood profiling over candidate values, following Engle et al. (2013). Model selection among nested specifications is guided by the Akaike Information Criterion (AIC), the Bayesian Information Criterion (BIC), and the likelihood ratio (LR) test. For the GARCH-MIDAS-X model, the author tests H_0: $\theta_X = 0$ against H_1: $\theta_X \neq 0$ using the LR statistic, which follows a $\chi^2(2)$ distribution under the null.

3.2 Data collection

Daily financial and monthly macroeconomic data are used to assess Bitcoin's hedging effectiveness against traditional assets under varying global economic policy uncertainty. The financial variables comprise Bitcoin, global equities, global bonds, and global commodities, proxied by the CoinDesk Bitcoin Price Index (XBX), the MSCI World Stock Index, the PIMCO Investment-Grade Corporate Bonds Index, and the S&P GSCI Commodity Index, respectively. To capture the macroeconomic environment, the Global Economic Policy Uncertainty (GEPU) index is included as the low-frequency explanatory variable. Specifically, data for Bitcoin is collected from CoinDesk, data for the conventional asset indices are directly obtained from LSEG Workspace, formerly Refinitiv Eikon to maintain consistency and accessibility, and the GEPU series is retrieved from the Economic Policy Uncertainty database (https://www.policyuncertainty.com). Beyond Bitcoin, the inclusion of equities, corporate bonds, and commodities aligns with empirical literature on portfolio optimization. This asset mix replicates a conventional global multi-asset portfolio, incorporating a broad spectrum of risk profiles to test Bitcoin's diversification benefits across distinct market segments. Given that this study evaluates global macroeconomic uncertainty, it is essential to incorporate asset classes that are inherently global in scope rather than country-specific.

The sample period spans from October 1, 2016 to December 31, 2025. This period is selected to capture different phases of market conditions and to ensure sufficient observations for estimating both the short-run and long-run components of volatility and correlation. Daily price series are used to compute return dynamics, while the monthly GEPU index is incorporated through the MIDAS structure to capture slower-moving macroeconomic effects on long-run variance and correlation.

In particular, the dataset includes the following variables:

- **Bitcoin Price:** Daily closing prices of the CoinDesk Bitcoin Price Index.
- **Global Equity Index:** Daily closing values of the MSCI World Stock Index.

- **Global Bond Index:** Daily closing values of the PIMCO Investment-Grade Corporate Bonds Index.
- **Global Commodity Index:** Daily closing values of the S&P GSCI Commodity Index.
- **GEPU Index:** Monthly values of the Global Economic Policy Uncertainty index obtained from the Economic Policy Uncertainty database.

Before model estimation, all financial market price series are transformed into logarithmic returns to ensure comparability across assets and to reduce scale effects. Since Bitcoin is traded continuously whereas conventional financial indices follow business-day trading calendars, the daily series are aligned to common trading dates prior to estimation. The monthly GEPU series is matched to the higher-frequency financial data through the MIDAS filtering structure. Missing observations associated with non-overlapping trading days, or market holidays are removed to maintain consistency in the estimation sample. This preprocessing step is particularly important for the DCC-MIDAS-X framework, which relies on synchronized return series to estimate dynamic conditional correlations and hedge ratios.

Summary of Chapter 3

This research evaluates, through a three-stage empirical framework, whether GEPU-conditioned hedge ratios sharpen Bitcoin's ability to protect a portfolio of global equities, commodities, and investment-grade bonds. The GARCH-MIDAS-X model first splits each asset's volatility into a short-run daily component and a long-run monthly component driven by realized volatility and GEPU, with Bai–Perron break dummies allowing that policy-uncertainty effect to shift across regimes rather than stay fixed. The DCC-MIDAS-X model then carries the same logic into Bitcoin's time-varying correlations with each conventional asset, separating transient co-movements from a slowly evolving long-run target shaped by realized correlations, GEPU, and the same regime shifts, before combining the estimated volatilities and correlations into the conditional covariance matrices that feed hedge-ratio construction. The framework is estimated by a two-step quasi-maximum-likelihood procedure with robust standard errors and full residual diagnostics over a sample running from October 2016 to December 2025. Crucially, the research turns the model into a tradeable hedging strategy, treating each conventional asset as the long position and Bitcoin as the hedging instrument to build optimal time-varying hedge ratios and then measure how much real portfolio risk they remove. That payoff is judged by variance reduction against the benchmark DCC-MIDAS model, tested in-sample and through rolling out-of-sample windows, confirmed with Clark–West and Diebold–Mariano statistics, and stress-checked against transaction costs of 10, 25, and 50 basis points, so that any demonstrated gain in Bitcoin's risk-reducing power survives the frictions a practitioner would actually face.

CHAPTER 4: EMPIRICAL RESULTS

4.1 Preliminary analysis of asset returns, static correlations, and GEPU structural breaks

Table 4.1: Summary statistics

Variable	Mean	Max	Min	Std.Dev	Skewness	Kurtosis	ADF test	JB test	PP test	ARCH(12)
Bitcoin	0.218	22.51	-46.47	4.3118	-0.5922	12.62	-12^{***}	8990^{***}	-2490^{***}	60^{***}
MSCI	0.0407	8.41	-10.44	0.9953	-1.3014	23.19	-13^{***}	39640^{***}	-2424^{***}	588^{***}
S&P GSCI	0.0183	7.68	-12.52	1.3799	-1.1501	13.35	-12^{***}	10749^{***}	-2258^{***}	228^{***}
PIMCO	0.0109	6.82	-5.08	0.4592	0.0012	44.08	-13^{***}	161424^{***}	-1993^{***}	754^{***}
GEPU	242.0946	623.35	125.99	81.8837	1.6744	7.09	-3^{***}	127.8^{***}	-30^{***}	61^{***}

Notes: JB stands for the Jarque-Bera test for normality. ADF and PP denote the Augmented Dickey-Fuller and Phillips-Perron unit root tests for stationarity, respectively. ARCH(12) represents Engle's Autoregressive Conditional Heteroskedasticity test conducted with 12 lags. The superscript "***" indicates statistical significance at the 1% level.

(Sources: Author's analysis, Refinitiv Eikon)

The sample covers October 2016 to December 2025 and includes five variables with sharply different risk profiles[6]. Bitcoin is the most volatile asset, with a daily standard deviation of 4.3118%, more than three times that of the S&P GSCI (1.3799%) and almost ten times that of PIMCO (0.4592%). This risk is matched by return. Bitcoin records the highest mean daily return at 0.218%, reflecting the risk-return profile that has attracted both speculative and institutional investors. By contrast, MSCI World and PIMCO report much lower mean returns of 0.04% and 0.01%, representing the more stable end of the sample. GEPU also shows substantial variation, with a mean of 242.09 and a maximum of 623.35. The wide gap between its minimum of 125.99 and maximum of 623.35 confirms the sharp shifts in global policy uncertainty across the four regimes examined in this study.

All four return series are non-normal, with negative skewness and excess kurtosis significant at the 1% level under the Jarque-Bera test. PIMCO records the highest kurtosis at 44.08, suggesting that even low-volatility investment-grade bonds are exposed to rare but severe tail events. Bitcoin and the S&P GSCI also show strong excess kurtosis, at 12.62 and 13.35 respectively, consistent with the fat-tailed behavior of high-risk assets. The ARCH(12) statistics confirm strong conditional

[6] Asset returns are reported in percentages, while GEPU is measured in index points.

heteroskedasticity across all return series, with the highest value for PIMCO (754) and the lowest for Bitcoin (60) which supports the use of a GARCH-based framework to model time-varying volatility. Finally, all series are stationary under both ADF and PP tests at the 1% level, confirming their suitability for direct estimation.

Table 4.2: Static correlation of Bitcoin, MSCI, S&P GSCI and PIMCO

	Bitcoin	MSCI	S&P GSCI	PIMCO
Bitcoin	1.000			
MSCI	0.277	1.000		
S&P GSCI	0.082	0.340	1.000	
PIMCO	0.101	0.288	0.015	1.000

(Sources: Author's analysis, Refinitiv Eikon)

Table 4.2 shows that Bitcoin's unconditional correlations with MSCI World (0.277), S&P GSCI (0.082), and PIMCO (0.101) are uniformly low, consistent with the diversification findings of Brière et al. (2015) and Bouri et al. (2017). However, these figures are full-sample averages that mask substantial regime-level heterogeneity. Bitcoin's correlation with MSCI World — already the highest of the three pairs at 0.277 — is well-documented to spike sharply during market stress episodes, as Conlon and McGee (2020) demonstrated during the COVID-19 crash. A hedging strategy based only on unconditional correlations will therefore be systematically inaccurate during high-uncertainty periods, when co-movement intensifies and hedging protection is most needed.

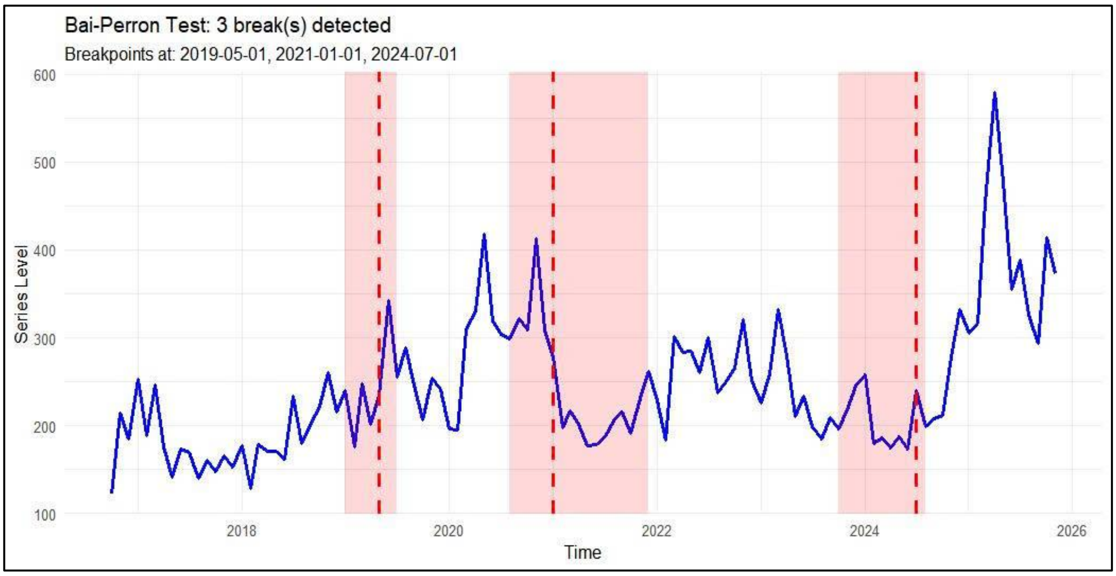
Table 4.3: Structural breaks for macroeconomic index (GEPU)

	Breakpoint 1	**Breakpoint 2**	**Breakpoint 3**
Break date	05/2019	01/2021	07/2024
BIC-selected breaks		3	
F-statistic		40.9577^{***}	
Scaled F-statistic		40.9577^{***}	
Weighted F-statistic		56.3528^{***}	

(Sources: Author's analysis)

Table 4.3 reports the results of the Bai and Perron (2003) sequential structural break test applied to the GEPU series. The procedure identifies three statistically significant break points — May 2019, January 2021, and July 2024 — selected by BIC criterion and supported by F-statistics that reject the null of parameter stability at the 1% level across all three test variants (F-statistic = 40.9577, Scaled F-statistic = 40.9577, Weighted F-statistic = 56.3528). These results confirm that global policy uncertainty has not evolved as a single stationary process over the sample period but has instead shifted discretely across four qualitatively distinct regimes.

Figure 4.2: Bai-Perron structural break test on GEPU index



(Sources: Author's analysis)

Regime 1 (October 2016 – April 2019) reflects the relatively contained uncertainty environment of synchronized global growth and gradual monetary policy normalization during the post-crisis recovery. World output strengthened to 3.8 percent in 2017, with a notable rebound in global trade and was projected at 3.9 percent in 2018 and 2019; the IMF (2018) noted that 120 countries, representing three-quarters of world economic output, enjoyed growth in 2017 — the broadest global expansion in seven years, while the Federal Reserve normalized policy through nine 25-bp hikes that lifted the funds rate to 2.25–2.50% by December 2018. GEPU fluctuated largely within a 130–250 band, reflecting a macro backdrop in which policy directions across major economies were broadly predictable and investor confidence was recovering (Baker et al., 2020).

The May 2019 break marks the sharp deterioration of US–China trade relations following the collapse of negotiations in early May 2019, when the Trump administration raised tariffs on $200 billion of Chinese goods from 10% to 25% and threatened to extend coverage to virtually all remaining Chinese imports (Office of the United States Trade Representative, 2019). The policy shock fundamentally altered the risk environment for globally exposed asset classes by injecting persistent uncertainty into trade policy, supply chain structures, and corporate investment plans in ways that Caldara et al. (2020) formally documented through their geopolitical risk index as a distinct and separately priced source of macroeconomic disruption.

Regime 2 (May 2019 – January 2021) captures the trade-war shock flowing directly into the acute pandemic phase. GEPU climbed to a regime peak of approximately 400–420 around the onset of COVID-19 in early 2020 — its highest level up to that point in the sample — as governments simultaneously shut down large portions of the global economy, deployed fiscal stimulus packages exceeding 10–15% of GDP in major economies, and central banks expanded balance sheets at unprecedented speed (Ahir, Bloom, and Furceri, 2022).

The January 2021 break marks the transition out of the acute pandemic phase and into the extraordinary policy response period. Regime 3 (January 2021 – June 2024) captures a fundamentally different but equally disruptive uncertainty environment, with GEPU settling into a roughly 200–300 range punctuated by the emergence of persistent inflation not seen since the 1980s, as US CPI peaked at 9.1% in June 2022 (U.S. Bureau of Labor Statistics, 2022), followed by the most aggressive monetary tightening cycle in four decades as the Federal Reserve raised the federal funds rate from near-zero to 5.25–5.50% across 2022–2023 (Federal Reserve, 2023). This regime is particularly consequential for the Bitcoin–PIMCO relationship, since rising rates both drove down bond prices and raised the opportunity cost of holding non-yielding assets like Bitcoin — a dual channel that likely tightened co-movement between the two asset classes.

The July 2024 break inaugurates the fourth and most structurally significant regime from the perspective of the present study. Regime 4 (July 2024 onward) records the highest GEPU readings in the entire sample, surging to approximately 580–600, and falls in the months following the SEC's approval of Bitcoin spot Exchange-Traded Funds in January 2024 — a decision that unlocked an estimated $10–15 billion in institutional capital inflows within the first quarter of approval and marked the formal integration of Bitcoin into the regulated financial system (U.S. Securities and Exchange Commission, 2024). The institutional entry fundamentally changed Bitcoin's investor composition, shifting from a predominantly retail and speculative base toward pension funds, sovereign wealth funds, and asset managers whose portfolio behavior is more directly conditioned on macroeconomic signals including GEPU (Fidelity Digital Assets, 2023). As a result, the GEPU transmission mechanism in this fourth regime is structurally different from all preceding periods — precisely the shift that the break dummy coefficients in the GARCH-MIDAS-X and DCC-MIDAS-X estimations are designed to capture.

4.2 Lag length selection

Table 4.4: GARCH-MIDAS lag length selection

Model	K (months)	Bitcoin		MSCI		S&P GSCI		PIMCO	
		LL	AIC	LL	AIC	LL	AIC	LL	AIC
Without GEPU (RV only)	6	-6452	12917	-2677	5367	-3749	7506	-769	1551
	12	-6452	12916	**-2673**	**5359**	**-3747**	**7505**	**-769**	**1551**
	24	**-6451**	**12915**	-2673	5359	-3748	7509	-770	1552
With GEPU (RV + GEPU)	6	-6448	12913	-2670	5357	-3745	7507	-765	1547
	12	**-6447**	**12910**	-2665	5347	**-3744**	**7503**	**-765**	**1546**
	24	-6447	12911	**-2664**	**5343**	-3745	7506	-768	1553
LR statistic ($\chi^2(2)$)	6	11.32^{***}		13.48^{***}		6.01^{***}		9.72^{***}	
	12	7.01^{***}		16.01^{***}		3.71^{**}		7.90^{**}	
	24	7.91^{***}		20.21^{***}		7.46^{***}		2.71^{**}	

Notes: LL and AIC stand for the Log-likelihood and Akaike Information Criterion. Bold and underline indicate the best specification in the GARCH-MIDAS estimation. The superscript "***", "**", "*" represent significance at the 1%, 5%, 10% level, respectively.

(Sources: Author's analysis)

The preferred lag length varies across assets and specifications, indicating that no single AIC-dominant window exists for all series. Under the GEPU specification, Bitcoin, S&P GSCI, and PIMCO achieve their best fit at the 12-month window, with AIC values of 12910, 7503, and 1546, respectively. MSCI World is the only exception, reaching its lowest AIC at the longest horizon of (K=24) months. This pattern suggests that most assets absorb realized variance and policy uncertainty through a one-year long-run component, while global equity volatility may respond to policy uncertainty over a more persistent horizon. For consistency across assets and to avoid asset-specific lag choices, this study adopts (K=12) months as the baseline MIDAS window, following the mixed-frequency modelling approach of Fang et al. (2019) and Xiong et al. (2024). Importantly, the significant LR statistics across all assets further support the inclusion of GEPU as an additional long-run volatility driver.

4.3 Univariate Volatility: GARCH-MIDAS-X results

Table 4.5: Estimation results of the baseline GARCH-MIDAS model (RV only) without structural breaks

	Bitcoin	**MSCI**	**S&P GSCI**	**PIMCO**
μ	0.2179***	0.0798***	0.0288**	0.0200***
	(3.4223)	*(5.8858)*	*(2.0917)*	*(3.2284)*
α	0.1491***	0.1550***	0.0946***	0.1017***
	(2.9766)	*(4.1565)*	*(2.6364)*	*(6.3249)*
β	0.7533***	0.8230***	0.8068***	0.8883***
	(16.4221)	*(15.9575)*	*(7.8299)*	*(52.8951)*
m	2.4135***	0.0416***	0.3390**	-1.6863***
	(10.8716)	*(6.1025)*	*(2.2333)*	*(-4.6303)*
θ_{RV}	4.5738***	1.7866***	-0.0041***	-0.0340
	(5.3287)	*(4.2825)*	*(-3.3772)*	*(-1.3657)*
ω_{RV}	6.1155***	5.6504***	29.9974***	3.5214**
	(3.0859)	*(52.1015)*	*(128.4078)*	*(1.9897)*
Log(L)	-6451.9	-2679.9	-3747.1	-758.4
AIC	12913.9	5369.8	7506.2	1528.8
BIC	12952.6	5398.6	7540.7	1563.3

Notes: Log(L), AIC, and BIC stand for Log-Likelihood, Akaike Information Criterion, and Bayesian Information Criterion, respectively. Standard errors are reported in parentheses. The superscripts ***, ** and * denote statistical significance at 1%, 5% and 10% levels, respectively.

(Sources: Author's analysis)

Table 4.5 presents the estimated results from the baseline GARCH-MIDAS model, in which realized variance is used as the only driver of the long-run volatility component. Most parameters in the return equation and the short-run variance component are statistically significant at the 1% level, indicating clear volatility clustering in daily asset returns. The sums of α and β are noticeably close to 1 for all four assets, especially MSCI and PIMCO, confirming strong volatility persistence effect and supporting the use of a GARCH-based specification to capture the slow adjustment of market risk.

The estimated θ_{RV} coefficients reveal clear cross-asset differences in how realized variance affects long-run volatility. For Bitcoin and MSCI World, θ_{RV} is positive and statistically significant at the 1% level, showing that higher past realized variance increases the long-run volatility component. This effect is especially strong for Bitcoin, suggesting that its persistent volatility is highly sensitive to past market turbulence. By contrast, S&P GSCI reports a small but significant negative coefficient, while PIMCO's coefficient is negative but insignificant. These results suggest that realized variance explains long-run volatility more clearly for Bitcoin and equities, whereas commodities and bonds display weaker or more mean-reverting long-run volatility dynamics under the RV-only specification.

Table 4.6: Estimation results of the baseline GARCH-MIDAS model (RV only) considering structural breaks

	Bitcoin	**MSCI**	**S&P GSCI**	**PIMCO**
μ	0.2404***	0.0766***	0.0300***	0.0201***
	(3.2038)	*(5.5808)*	*(4.1733)*	*(3.3204)*
α	0.1506***	0.1355***	0.0825***	0.1353***
	(2.9141)	*(5.2188)*	*(3.2976)*	*(5.3946)*
β	0.7307***	0.8475***	0.8535***	0.7410***
	(7.6455)	*(17.4321)*	*(14.6333)*	*(13.9774)*
m	2.6316***	-0.4531***	0.6512***	-2.5696***
	(12.2878)	*(-13.2714)*	*(3.3418)*	*(-17.5071)*
θ_{RV}	4.2087***	1.0708***	-0.0191**	-0.1511
	(5.3271)	*(3.1967)*	*(-2.4049)*	*(-0.9856)*
ω_{RV}	8.2415***	1.0010**	29.9808***	9.1547***
	(3.7812)	*(2.4625)*	*(80.5484)*	*(2.6506)*
$\theta_{RV,\ break1}$	-1.7769***	1.2073***	0.0228***	0.1706
	(-8.4025)	*(3.5964)*	*(2.8555)*	*(1.1507)*
$\theta_{RV,\ break2}$	1.0001***	-1.0835	0.0185***	0.3717
	(4.5491)	*(-0.0006)*	*(2.6266)*	*(0.6991)*

$\theta_{RV,\ break3}$	9.6082***	4.6157***	0.0029	0.3349**
	(36.9130)	*(44.2607)*	*(0.3264)*	*(2.3410)*
Log(L)	-6434.9	-2669.8	-3731.6	-743.1
AIC	12891.8	5355.6	7481.3	1504.1
BIC	12904.9	5390.9	7532.9	1555.7

Notes: See note of Table 4.5

(Sources: Author's analysis)

Table 4.6 extends the RV-only benchmark by allowing the θ_{RV} coefficient to shift across the Bai–Perron break regimes. The lower AIC and BIC values across all four assets, together with the likelihood-ratio tests in Appendix C, Panel A, indicate that the structural-break specification fits the data better than the single-regime model. This confirms that the relationship between realized variance and long-run volatility is not stable over time.

The break coefficients further show that this instability is concentrated in Bitcoin, MSCI, and S&P GSCI, while PIMCO remains largely insensitive to realized variance across regimes. For Bitcoin, $\theta_{RV,break1}$ is negative and significant, indicating that the trade war period weakened the link between realized variance and long-run volatility — consistent with a market still driven by speculative sentiment where fundamental risk signals had limited traction. This relationship turned positive and significant in the pandemic regime (break 2) and amplified further in the post-ETF period (break 3), reflecting Bitcoin's progressive institutional maturation across successive regimes. For MSCI World, break 1 and break 3 are both positive and significant, suggesting that global equity long-run volatility became increasingly sensitive to realized variance as policy uncertainty intensified over time. For S&P GSCI, breaks 1 and 2 are positively significant while break 3 is not, indicating that commodity volatility experienced regime shifts during the trade war and pandemic periods but remained unaffected by the post-ETF structural change — consistent with a volatility process still governed by supply-demand fundamentals rather than macro-financial integration (IMF, 2024). And for PIMCO, all break coefficients are insignificant, confirming that investment-grade bond volatility is insensitive to realized variance across all regimes. Overall, these results show that the RV–volatility relationship is structurally unstable for most assets, so a single-regime specification would mask important regime-specific volatility dynamics.

Table 4.7: Estimation results of the GARCH-MIDAS-X model (RV and GEPU) without structural breaks

	Bitcoin	**MSCI**	**S&P GSCI**	**PIMCO**
μ	0.2477^{***}	0.0835^{***}	0.0311^{**}	0.0202^{***}
	(3.2320)	*(5.8654)*	*(2.2444)*	*(3.2864)*
α	0.1466^{***}	0.1775^{***}	0.0878^{***}	0.1038^{***}
	(3.2284)	*(4.7093)*	*(3.2791)*	*(4.6487)*
β	0.7418^{***}	0.7908^{***}	0.8506^{***}	0.8863^{***}
	(15.7341)	*(20.7549)*	*(16.9229)*	*(34.6372)*
m	2.4719^{***}	-0.6549^{***}	0.4831^{***}	-1.6666^{***}
	(9.4917)	*(-2.6092)*	*(3.3359)*	*(-5.0248)*
θ_{RV}	7.5241^{***}	0.0275^{***}	0.0047^{*}	-0.0490^{**}
	(18.8066)	*(2.6824)*	*(1.7237)*	*(-2.0804)*
ω_{RV}	7.4545^{***}	1.0010^{**}	1.3137	3.0250^{***}
	(9.9032)	*(2.3601)*	*(0.7551)*	*(2.8266)*
θ_X	-6.0322^{***}	2.2067^{**}	3.1204^{***}	6.8296^{**}
	(-3.8057)	*(2.1467)*	*(3.2733)*	*(2.1939)*
ω_X	2.2873^{***}	3.6487^{**}	4.5321^{***}	1.0010^{***}
	(2.6995)	*(2.3507)*	*(4.7381)*	*(3.0331)*
Log(L)	-6448.1	-2652.1	-3745.1	-754.6
AIC	12910.1	5320.3	7506.2	1525.3
BIC	12950.3	5366.2	7552.1	1571.2

Notes: See note of Table 4.5

(Sources: Author's analysis)

Table 4.7 incorporates the long-run volatility equation with GEPU as an additional conditioning variable alongside realized variance. The short-run GARCH parameters remain stable and consistent with Table 4.5, confirming that the introduction of GEPU does not distort the short-run volatility dynamics. The key new evidence lies in the θ_X coefficients, which measures the impact of macroeconomic shocks on the long-term volatility of Bitcoin, MSCI World, S&P GSCI, and PIMCO.

Before turning to the GEPU coefficients, the signs of the θ_{RV} estimates are informative. For Bitcoin and MSCI World, θ_{RV} is positively significant— higher realized variance feeds into higher long-run volatility, the typical pattern for risk assets — while for PIMCO it stays negative, consistent with mean-reverting bond volatility. The key case is S&P GSCI, whose θ_{RV} flips from negative in Table 4.5 to positive once GEPU is added: the apparent mean-reversion under the RV-only model was in fact absorbing the influence of policy uncertainty, and controlling for GEPU restores the expected positive RV relationship.

The sign and significance of θ_X differ sharply across asset classes, revealing fundamentally different transmission mechanisms. For Bitcoin, $\theta_X = -6.0322$ is negatively significant, indicating that rising GEPU leads to lower long-term volatility component — a result that appears counterintuitive but is consistent with the flight-from-risk narrative: during high-uncertainty periods, speculative activity in Bitcoin contracts as investors reduce risk exposure, compressing the persistent volatility component while short-run GARCH dynamics remain unaffected. In contrast to Fang et al. (2019), who documented a positive θ_X for Bitcoin in their 2010–2018 sample, the negative coefficient estimated here reflects Bitcoin's structural transformation following institutional maturation: as macro-oriented investors came to dominate the market — particularly after the SEC's approval of spot ETFs in January 2024 — rising GEPU now triggers systematic risk-off reductions in Bitcoin exposure rather than the speculative volatility amplification that characterized the earlier period. For MSCI World and S&P GSCI, θ_X is positively significant (2.2067 and 3.1204 respectively), consistent with the asset-specific transmission channels described by Arouxet et al. (2024), whereby rising policy uncertainty increases risk premia and feeds into higher persistent volatility for risk-sensitive assets. For PIMCO, $\theta_X = 6.8296$ is positive and significant, reflecting the amplified duration and credit spread sensitivity of investment-grade bonds during episodes of monetary policy uncertainty — where ambiguity about the future rate path translates directly into elevated long-run bond return volatility. Comparing AIC and BIC with Table 4.5, the GEPU-augmented specification produces lower information criteria for Bitcoin and MSCI World, confirming that GEPU carries incremental explanatory power for long-run volatility beyond realized variance alone for these two assets.

Beyond the sign of θ_X, the MIDAS weighting parameter ω_X shows that GEPU is absorbed at different speeds across asset classes. Bitcoin has the lowest ω_X, suggesting a faster response to recent policy-uncertainty shocks, while S&P GSCI has the highest value, implying a slower and more distributed transmission consistent with commodity adjustment lags. However, the limited AIC and BIC improvement for S&P GSCI and PIMCO suggests that a constant GEPU effect may still understate the role of policy uncertainty for these assets.

Table 4.8: Estimation results of the GARCH-MIDAS-X model (RV and GEPU) considering structural breaks

	Bitcoin	MSCI	S&P GSCI	PIMCO
μ	0.2474***	0.0763***	0.0408*	0.0211***
	(3.3561)	*(5.6703)*	*(1.7315)*	*(3.5478)*
α	0.1391***	0.1488***	0.0804***	0.1334***
	(3.6636)	*(5.3368)*	*(4.2478)*	*(5.2782)*
β	0.7310***	0.8135***	0.8840***	0.7357***
	(16.5465)	*(26.3880)*	*(30.3583)*	*(14.6475)*
m	2.6307***	-0.5386***	0.6230**	-2.5271***
	(10.2113)	*(-3.7990)*	*(2.4796)*	*(-17.2682)*
θ_{RV}	8.1647***	0.0664**	0.0153***	-0.2183**
	(3.4448)	*(2.3414)*	*(5.7258)*	*(-2.3493)*
ω_{RV}	11.4380***	1.0010***	27.8495***	3.0947*
	(7.4801)	*(4.0114)*	*(16.3391)*	*(1.9167)*
θ_X	-5.3840***	1.6122*	2.6019**	1.1892**
	(-2.6582)	*(1.7110)*	*(2.1616)*	*(2.3940)*
ω_X	3.3342***	29.9241***	2.3879***	2.6286***
	(4.1826)	*(8.3732)*	*(3.6691)*	*(4.2097)*
$\theta_{RV,\ break1}$	-4.1247	0.0812***	0.0203**	0.2491
	(-0.6013)	*(3.2274)*	*(2.4855)*	*(1.6000)*
$\theta_{RV,\ break2}$	-2.7368	0.0720***	0.0139	0.4341***
	(-0.5446)	*(3.3227)*	*(1.5591)*	*(2.9920)*
$\theta_{RV,\ break3}$	-8.1323***	0.0893***	-0.0167*	0.2279*
	(-5.3277)	*(3.4548)*	*(-1.9130)*	*(1.7382)*
$\theta_{X,\ break1}$	-8.5821	-1.2733	-0.4669	-5.3390*
	(-0.6892)	*(-1.2873)*	*(-1.3748)*	*(-1.8104)*
$\theta_{X,\ break2}$	2.8551***	0.4878	-0.4276**	3.4109
	(11.8609)	*(0.4625)*	*(-2.5160)*	*(0.9155)*

$\theta_{X,\ break3}$	0.0780	5.5725***	1.4885***	7.7693***
	(0.1510)	*(5.5182)*	*(3.4337)*	*(2.6974)*
Log(L)	-6438.0	-2607.8	-3708.9	-731.1
AIC	12904.0	5243.7	7445.7	1490.3
BIC	12944.3	5324.1	7526.1	1570.6

Notes: See note of Table 4.5

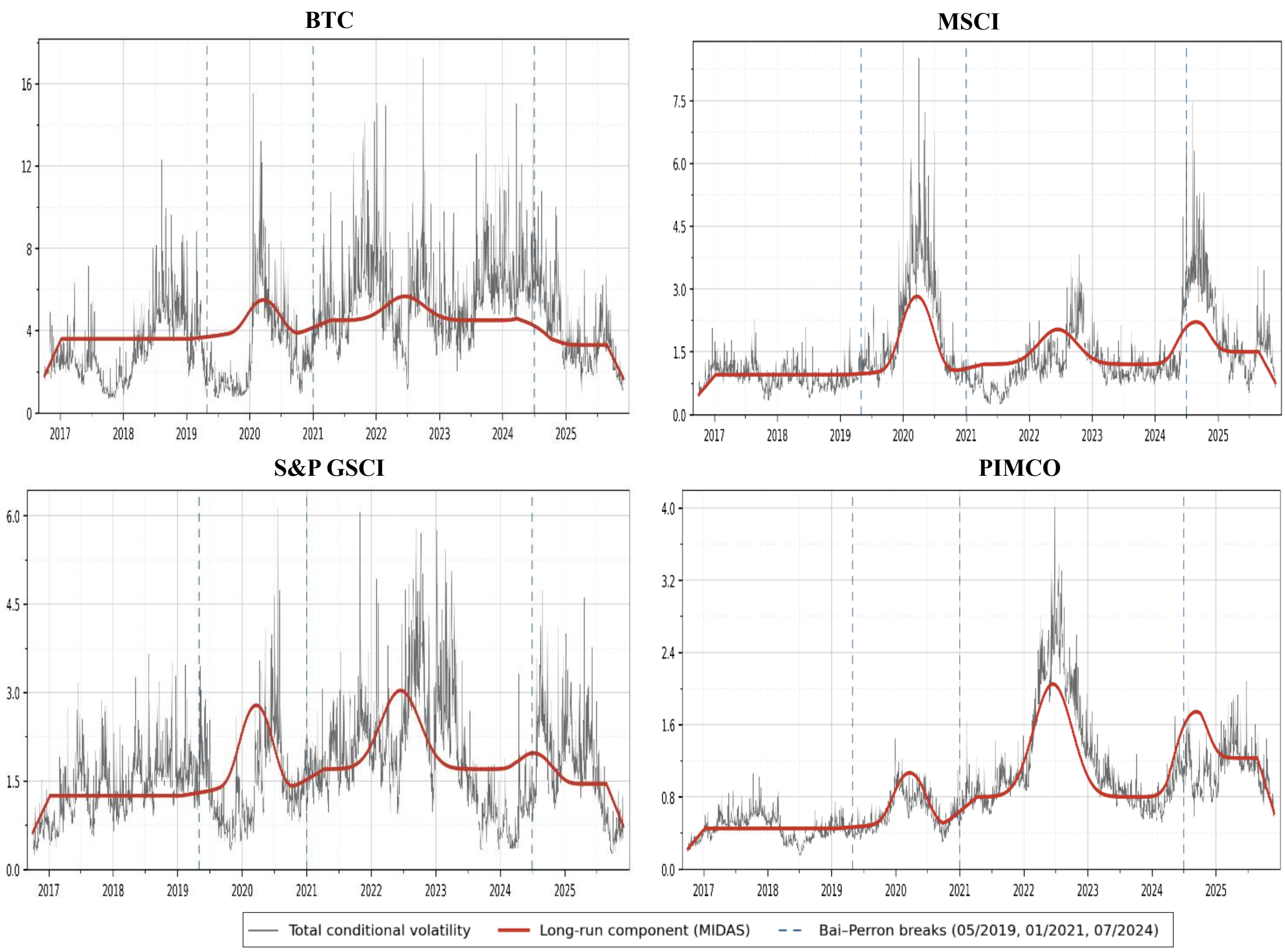
(Sources: Author's analysis)

Table 4.8 extends the single-regime GARCH-MIDAS-X specification in Table 4.7 by allowing both θ_{RV} and θ_X to shift across the three Bai–Perron regimes. Compared with Table 4.7, the lower AIC and BIC values across all four assets confirm that incorporating structural breaks improves the univariate model fit over the full sample.

The θ_X break coefficients show that GEPU transmission is strongly regime-dependent. For Bitcoin, the baseline θ_X remains negative (−5.3840), consistent with Table 4.7, but the break coefficients show meaningful regime shifts: $\theta_{X,break2} = 2.8551$ is positively significant, indicating that during the pandemic and early recovery period GEPU's suppressive effect on Bitcoin's long-run volatility reversed — rising uncertainty during this regime amplified rather than compressed Bitcoin's persistent volatility component, likely reflecting the surge in retail and institutional speculative interest that characterized the 2020–2021 crypto bull market. $\theta_{X,break3}$ is essentially zero and insignificant, suggesting that in the post-ETF regime the direct GEPU–Bitcoin volatility relationship stabilized as institutional investors priced policy uncertainty more systematically. For MSCI World, $\theta_{X,break3} = 5.5725$ is large and significant, indicating a dramatic amplification of GEPU's effect on global equity long-run volatility in the post-ETF regime — consistent with the broader re-pricing of macro risk that accompanied the normalization of monetary policy after 2024. For S&P GSCI, $\theta_{X,break2} = -0.4276$ is negative and significant while $\theta_{X,break3} = 1.4885$ is positive and significant, suggesting that commodity volatility's sensitivity to GEPU switched direction across regimes — from a dampening effect during the pandemic supply shock period to an amplifying effect in the most recent regime as energy and commodity markets became more integrated with global macro conditions. For PIMCO, $\theta_{X,break3} = 7.7693$ is the largest GEPU break coefficient in the table and highly significant, capturing the extraordinary sensitivity of investment-grade bond volatility to policy uncertainty in the post-ETF regime — a period defined by the highest real interest rate uncertainty in four decades as central banks navigated the transition from emergency accommodation back to restrictive policy.

Overall, Table 4.8 provides strong support for H1 and H3 at the univariate stage. GEPU significantly affects long-run volatility, but its impact is not stable across regimes or asset classes. Therefore, a single-regime GARCH-MIDAS-X model would average across structurally different uncertainty environments and misstate the true policy-uncertainty channel. Having established this asset-level evidence, the analysis now turns to the bivariate setting in Section 4.4, where the key question is whether GEPU also shapes Bitcoin's long-run correlations with conventional assets and improves the hedge ratios evaluated in Section 4.5.

Figure 4.3: Conditional and long-run MIDAS volatility with structural breaks



. *(Sources: Author's analysis)*

4.4 Dynamic Conditional Correlation: DCC-MIDAS-X results

Table 4.9: Estimation results of the baseline DCC-MIDAS model for Bitcoin-asset pairs with and without structural breaks

	Without structural breaks			**Considering structural breaks**		
	BTC-MSCI	*BTC-GSCI*	*BTC-PIMCO*	*BTC-MSCI*	*BTC-GSCI*	*BTC-PIMCO*
a	0.0013	0.0093	0.0069	0.0107	*0.0089*	0.0038
	(1.2182)	*(1.9098)*	*(0.4561)*	*(0.7593)*	*(1.7014)*	*(0.6914)*
b	0.9905***	0.9867***	0.9877***	0.9834***	*0.9722****	0.9746***
	(3.3186)	*(155.2289)*	*(58.1985)*	*(65.4479)*	*(46.5983)*	*(37.7100)*
θ_{RC}	0.6078***	-1.6448***	-0.8872	0.3369***	*-1.7497****	-0.9259
	(4.3506)	*(-42.0910)*	*(-0.6137)*	*(38.3912)*	*(-5.8256)*	*(-0.3223)*
$\theta_{RC,\ break1}$				0.3282***	*1.9235****	1.1140
				(30.0965)	*(2.6920)*	*(0.4211)*
$\theta_{RC,\ break2}$				0.3327***	*1.6475***	1.3642
				(38.8642)	*(2.5711)*	*(0.4788)*
$\theta_{RC,\ break3}$				-0.3076***	*-0.2432*	-2.0004*
				(-8.1207)	*(-0.3201)*	*(-1.6739)*
Log(L)	-4407.8	-4507.7	-4567.2	-4373.9	-4479.2	-4488.4
AIC	8825.7	9025.4	9144.4	8769.9	8974.3	8992.8
BIC	8854.4	9054.1	9173.1	8833.0	9020.2	9038.7

Notes: See note of Table 4.5

(Sources: Author's analysis)

Table 4.9 reports the baseline DCC-MIDAS results for the three Bitcoin–asset pairs using realized correlations only. Across all specifications, the short-run shock parameter a is insignificant, while the persistence parameter b is close to 1 and highly significant. This indicates that Bitcoin's dependence with conventional assets is driven mainly by persistent long-run correlation dynamics rather than temporary daily shocks.

For the parameter estimates of θ_{RC}, the results differ markedly across asset pairs and carry distinct economic interpretations. θ_{RC} is significantly positive for

BTC-MSCI (0.6078), indicating that periods of elevated realized co-movement between Bitcoin and global equities feed persistently into the long-run correlation target. This reflects Bitcoin's tendency to behave as a high-beta risk asset whose co-movement with equities is self-reinforcing — when investors treat Bitcoin as part of the broader risk-on trade, that behavior accumulates into a structurally higher long-run correlation. Accordingly, investors can consider Bitcoin as a hedging instrument for MSCI during periods of negative realized Bitcoin–equity co-movement, when the two assets are moving in opposite directions. For BTC-GSCI, θ_{RC} is significantly negative (−1.6448), implying that higher realized Bitcoin–commodity correlation is associated with mean-reversion rather than amplification in the long-run target. Economically, this reflects the transitory nature of Bitcoin–commodity co-movement episodes, which are typically driven by short-run energy cost linkages tied to Bitcoin mining activity rather than by persistent macro-financial integration — when these short-run forces dissipate, the long-run correlation reverts. This suggests that Bitcoin can serve as a hedging instrument for S&P GSCI precisely during periods of positive realized co-movement, as the mean-reverting dynamic implies that elevated co-movement is temporary and followed by divergence. For BTC-PIMCO, θ_{RC} is negative but insignificant, indicating that past realized correlation carries no reliable information about the long-run Bitcoin–bond relationship whatsoever. This finding is consistent with the flight-to-quality literature—the Bitcoin–bond long-run correlation is not driven by historical co-movement patterns but by the prevailing macroeconomic uncertainty environment, which simultaneously pushes capital into safe-haven bonds while its effect on Bitcoin depends on whether investors perceive it as a risk asset or an alternative store of value.

Introducing structural breaks produces lower AIC and BIC values across all three pairs — most substantially for BTC-MSCI (8,825.7 to 8,769.9) and BTC-PIMCO (9,144.4 to 8,992.8) — confirming that the structural-break specification fits the data better than the single-regime baseline. This confirms that realized correlation transmission is not stable over the full sample and is better captured when regime shifts across the four GEPU periods are explicitly allowed. The break coefficients reveal economically meaningful shifts tied directly to the macroeconomic events that define each regime.

For BTC-MSCI, breaks 1 and 2 are both positive and significant (0.3282 and 0.3327), indicating that both the trade war escalation of May 2019 and the COVID-19 shock of January 2021 produced persistent upward shifts in the long-run Bitcoin–equity correlation — consistent with the risk-off contagion mechanism documented by Akhtaruzzaman et al. (2021), where investors simultaneously liquidated both Bitcoin and equities during acute stress episodes, generating synchronized drawdowns that permanently elevated the long-run co-movement target. Break 3 then turns negative and significant (−0.3076), suggesting that the post-ETF institutional regime beginning July 2024 partially reversed this dynamic, as the entry of macro-oriented institutional investors who manage Bitcoin as a distinct asset class decoupled it from the retail-driven equity co-movement that characterized earlier regimes.

For BTC–GSCI, the large positive and significant breaks 1 and 2 (1.9235 and 1.6475) reflect the synchronized commodity and crypto demand shocks during the trade war and pandemic periods — when supply chain disruptions and energy market dislocations created correlated volatility across Bitcoin and commodity markets — before the relationship stabilized with an insignificant break 3 as energy markets normalized and cross-market stress pressures gradually faded.

For BTC-PIMCO, the single significant break is break 3 (−2.0004), capturing a sharp structural decline in the long-run Bitcoin–bond correlation in the post-ETF regime. This finding is consistent with the flight-to-quality mechanism becoming more operative as institutional investors increasingly allocated to both Bitcoin and investment-grade bonds as complementary portfolio instruments — Bitcoin for asymmetric upside exposure and bonds for income stability — rather than treating them as competing claims on the same pool of risk capital. Taken together, the significance of break coefficients across all three pairs confirms that a single-regime DCC-MIDAS produces systematically biased long-run correlation estimates, and that the economic relationships between Bitcoin and conventional assets are fundamentally shaped by the prevailing uncertainty regime rather than being stable structural parameters. These findings constitute preliminary support for H3 in the correlation dimension, and directly motivate the extension to the DCC-MIDAS-X specification in Table 4.10, which introduces GEPU as an explicit conditioning variable to provide the formal test of H2.

Table 4.10: Estimation results of the DCC-MIDAS-X model for Bitcoin-asset pairs with and without structural breaks

	Without structural breaks			**Considering structural breaks**		
	BTC-MSCI	*BTC-GSCI*	*BTC-PIMCO*	*BTC-MSCI*	*BTC-GSCI*	*BTC-PIMCO*
a	0.0189	0.0092	0.0056	0.0147	*0.0091*	0.0051
	(0.9026)	*(0.8027)*	*(0.9894)*	*(0.4033)*	*(0.3563)*	*(0.0209)*
b	0.9771***	0.9866***	0.9918***	0.9809***	*0.9857****	0.9931***
	(154.9168)	*(63.4400)*	*(138.9519)*	*(146.8178)*	*(36.2986)*	*(3.6277)*
ω_{RC}	1.001**	2.4295***	6.6846***	8.6172***	*1.0213**	4.3887***
	(2.4766)	*(4.1881)*	*(24.6190)*	*(6.1563)*	*(1.9510)*	*(7.1208)*
θ_{RC}	-3.2403**	-4.3959**	-5.4366***	-1.1079**	*-18.6255****	-4.6398***
	(-2.4951)	*(-2.4838)*	*(-16.5855)*	*(-2.4313)*	*(-9.6707)*	*(-4.8000)*
θ_X	-1.7977***	0.7068**	0.4709***	-1.5051***	*0.6847***	0.4495***
	(-15.0677)	*(2.0272)*	*(9.8127)*	*(-3.6453)*	*(2.0141)*	*(7.4534)*
ω_X	11.6793***	4.4683***	2.2902***	25.9935***	*3.5215****	1.0010
	(3.3439)	*(86.6263)*	*(50.2735)*	*(20.0308)*	*(26.0936)*	*(1.5547)*
$\theta_{RC,\ break1}$				-0.7413	*15.4994****	1.9302*
				(-0.6050)	*(34.3240)*	*(1.7436)*
$\theta_{RC,\ break2}$				-0.2501	*6.1042****	4.6086***
				(-0.2944)	*(6.4658)*	*(7.9673)*
$\theta_{RC,\ break3}$				1.1568***	*4.7416*	-0.7631
				(29.5419)	*(0.0207)*	*(-0.5366)*
$\theta_{X,\ break1}$				1.3208***	*-2.2190*	-3.3976***
				(16.5492)	*(-0.0466)*	*(-5.9610)*

$\theta_{X,\ break2}$				1.2056^{*}	*0.6488^{***}*	4.4839^{***}
				(1.6521)	*(54.6351)*	*(3.9447)*
$\theta_{X,\ break3}$				-1.1367	*1.4565*	0.1434
				(-0.9369)	*(0.0307)*	*(0.3424)*
Log(L)	-4379.1	-4505.2	-4584.9	-4340.7	-4478.7	-4463.6
AIC	8772.1	9024.5	9183.9	8707.4	8973.3	8953.2
BIC	8812.2	9064.7	9224.1	8782.0	9057.9	9027.8

Notes: See note of Table 4.5

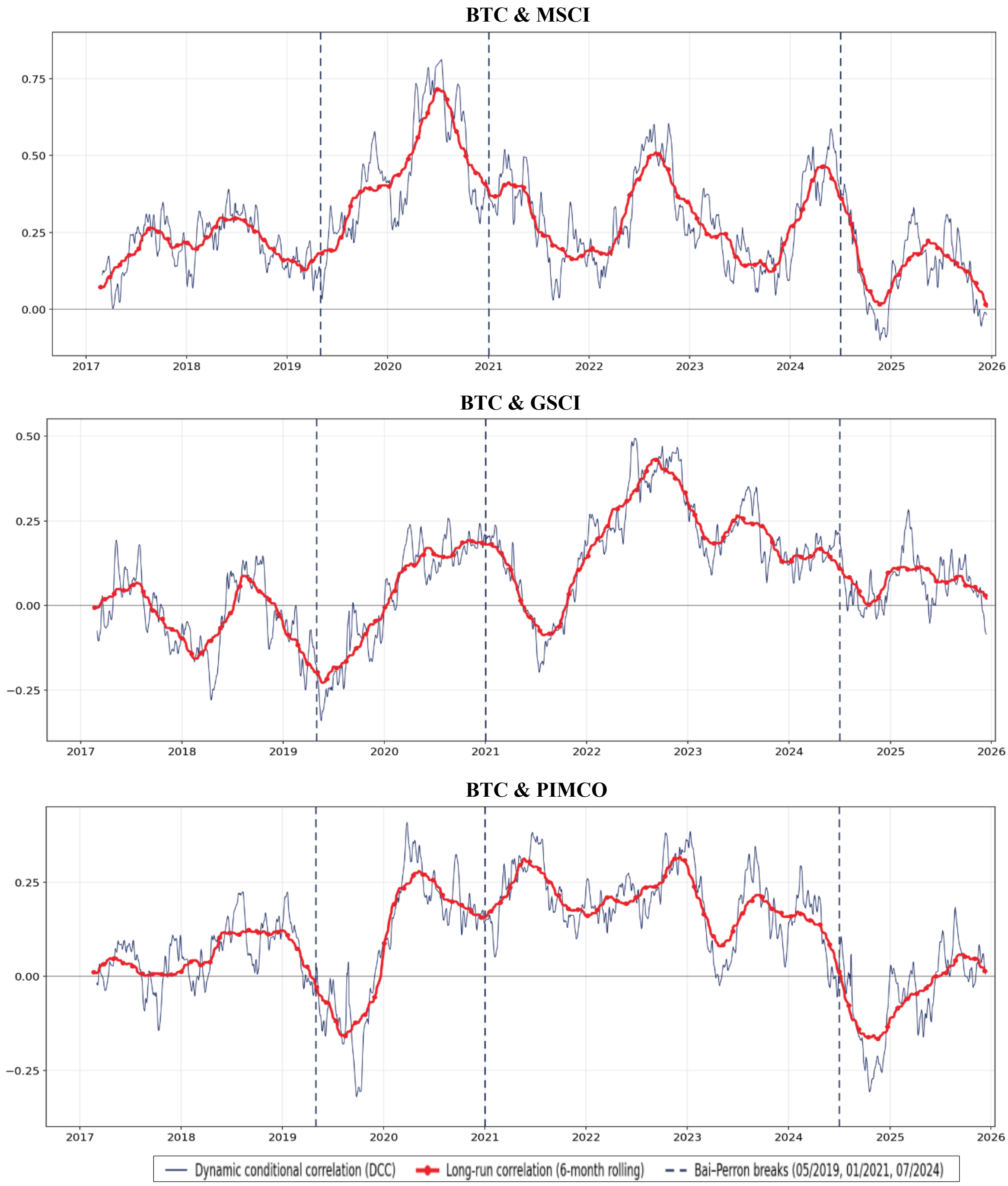
(Sources: Author's analysis)

Table 4.10 extends the correlation framework by adding GEPU as a direct conditioning variable in the long-run correlation equation, with and without structural breaks. The persistence parameter *b* remains close to 1 and highly significant across all pairs and specifications, consistent with Table 4.9. The baseline θ_{RC} coefficients are now negatively significant across all three pairs — in contrast to the mixed signs in Table 4.9 — indicating that once GEPU is controlled for, higher realized correlation is associated with mean-reversion in the long-run correlation target rather than amplification, suggesting that GEPU absorbs the persistent macro-driven component of co-movement that realized correlation alone was previously capturing.

The θ_X coefficients carry the central economic content of this table. For BTC-MSCI, $\theta_X = -1.7977$ is negative and significant, indicating that rising GEPU reduces the long-run Bitcoin–equity correlation — meaning that Bitcoin's co-movement with global equities actually declines during high-uncertainty episodes, consistent with Bitcoin becoming less sensitive to equity market movements during periods of elevated macro uncertainty. From a portfolio management perspective, this finding implies that Bitcoin becomes more useful as an equity hedge when policy uncertainty rises — the exact episodes where institutional investors most urgently require effective risk reduction tools. For BTC-GSCI and BTC-PIMCO, θ_X is positive and significant (0.7068 and 0.4709 respectively), indicating that rising GEPU increases Bitcoin's long-run co-movement with both commodities and bonds. This means that during high-uncertainty periods, Bitcoin, commodities, and investment-grade bonds tend to move more closely together — likely because all three asset classes are simultaneously exposed to the same macro shocks, including demand uncertainty for commodities and monetary policy ambiguity for bonds. Practically, Bitcoin becomes a less effective hedge for both S&P GSCI and PIMCO, as its diversification benefits diminish precisely when macro conditions are most disruptive and intense. The sign asymmetry across asset classes directly confirms the asset-class heterogeneity in GEPU transmission hypothesized in H2 and documented in the literature by Fang et al. (2019).

Introducing structural breaks again produces uniformly lower AIC and BIC across all three pairs — most substantially for BTC-MSCI (AIC falls from 8,772.1 to 8,707.4) — confirming that the GEPU–correlation relationship is regime-dependent and that a single-regime DCC-MIDAS-X misrepresents the long-run correlation dynamics across the four uncertainty regimes. The break coefficients reveal that GEPU's influence on long-run correlations shifted meaningfully across regimes. For BTC-MSCI, $\theta_{X,break1}$ and $\theta_{X,break2}$ are both positive and significant (1.3208 and 1.2056), indicating that during the US-China trade war and COVID-19 pandemic regimes, rising GEPU actually strengthened Bitcoin–equity co-movement — temporarily overriding the negative baseline θ_X and suggesting that in these early regimes Bitcoin still behaved as a high-beta risk asset during uncertainty spikes. $\theta_{X,break3}$ is insignificant, indicating that this dynamic stabilized in the post-ETF regime. For BTC-PIMCO, the break pattern is striking: $\theta_{X,break1} = -3.3976$ is large, negative, and significant, meaning that during the trade war period rising GEPU strongly reduced Bitcoin–bond co-movement — consistent with the flight-to-quality mechanism operating most powerfully during the first major uncertainty regime — while $\theta_{X,break2} = 4.4839$ reverses this effect in the pandemic period as both Bitcoin and bonds were simultaneously affected by the extraordinary policy response. For BTC-GSCI, $\theta_{X,break2} = 0.6488$ is positively significant while breaks 1 and 3 are insignificant, suggesting that GEPU's amplifying effect on Bitcoin–commodity co-movement was concentrated in the pandemic regime and did not persist into the post-tightening environment. Across all three pairs, the joint significance of the break coefficients confirms that GEPU's role in shaping long-run Bitcoin correlations is fundamentally regime-dependent, and that the single-regime estimates of the no-break specification produce systematically biased correlation and hedge ratio estimates across the full sample period.

Figure 4.4: Dynamic and long-run correlations between Bitcoin and conventional asset indices with structural breaks



(Sources: Author's analysis)

4.5 Hedging effectiveness

Table 4.11: Comparisons of hedging performance of Bitcoin across different asset indices and model specifications

Model	BTC–MSCI	BTC–GSCI	BTC–PIMCO
	HE	*HE*	*HE*
DCC-RC (no SB)	0.1616	0.0176	0.0220
DCC-RC (with SB)	0.1698	0.0184	0.0301
DCC-X (no SB)	0.1628	0.0147	0.0197
DCC-X (with SB)	**0.1705**	**0.0218**	**0.0303**
	ΔHE	*ΔHE*	*ΔHE*
Δ (GEPU effect)	+0.0013	−0.0029	−0.0024
Δ (SB effect)	+0.0083	−0.0053	+0.0081
Δ (full extension)	**+0.0089**	**+0.0042**	**+0.0082**

(Sources: Author's analysis)

To assess whether GEPU conditioning and structural break accommodation translate into tangible portfolio benefits, the author substitutes the conditional covariance matrix estimated from each of the four model specifications — DCC-RC (no SB), DCC-RC (with SB), DCC-X (no SB), and DCC-X (with SB) — into Equation (18) and computes the optimal variance-minimizing hedge ratio for each Bitcoin–asset pair. The resulting hedging effectiveness values, reported in Table 4.11, allow a direct comparison of portfolio variance reduction across specifications and a clean decomposition of the incremental gains attributable to GEPU conditioning and structural break accommodation respectively.

The full DCC-MIDAS-X with structural breaks delivers the highest HE across all three pairs, with BTC-MSCI achieving the strongest variance reduction at 0.1705, followed by BTC-PIMCO (0.0303) and BTC-GSCI (0.0218). These figures confirm that incorporating both GEPU conditioning and regime-aware parameter estimation consistently produces the most accurate hedge ratios across all asset classes.

However, the absolute magnitudes remain modest — peaking at 17.05% variance reduction for the equity pair — reflecting the fundamental constraint imposed by Bitcoin's extreme volatility on the achievable scale of portfolio risk reduction. As argued by Bouri et al. (2018), Bitcoin's price dynamics are considerably driven by network sentiment and speculative behavior rather than macroeconomic fundamentals, which limits the extent to which any macro-conditioned model can fully harness its hedging potential. Crucially, the improvement is not only economically meaningful but also statistically supported. The Clark–West and Diebold–Mariano tests in Appendix C, Panel B reject equal predictive accuracy in favour of the full model, confirming that the superior hedging performance reflects genuine forecasting gains rather than incidental in-sample improvement.

The decomposition of ΔHE reveals that GEPU conditioning and structural break accommodation contribute differently across asset classes. For BTC-MSCI, the structural break effect (+0.0083) dominates the GEPU effect (+0.0013), indicating that regime awareness — rather than macro conditioning per se — is the primary driver of hedging improvement for the equity pair. This is consistent with the finding that Bitcoin–equity co-movement is fundamentally regime-dependent, and that correctly identifying regime transitions matters more than the level of GEPU itself for constructing accurate equity hedge ratios. For BTC-PIMCO, the pattern is similar — structural breaks contribute +0.0081 while GEPU alone yields −0.0024 — implying that GEPU conditioning without structural break accommodation actually degrades hedging effectiveness for the bond pair. This occurs because a single-regime GEPU coefficient averages across structurally different periods where GEPU's effect on the Bitcoin–bond correlation operates in opposite directions, producing a miscalibrated hedge ratio that is worse than the realized correlation baseline. The full extension recovers this loss and delivers a net improvement of +0.0082, confirming that for investment-grade bonds it is the interaction between GEPU conditioning and regime awareness that generates genuine hedging value — consistent with the flight-to-quality mechanism operating most powerfully during specific uncertainty regimes rather than uniformly across the full sample.

The BTC–GSCI pair presents the most instructive case. Applied in isolation, neither GEPU conditioning nor structural break accommodation improves hedging for the commodity pair — each extension independently generates small negative ΔHE. The positive full-extension gain (+0.0042) therefore arises entirely from their interaction: as Table 4.10 shows, GEPU's effect on the Bitcoin–commodity long-run correlation is concentrated in a single regime (pandemic period, $\theta_{X,\text{ break2}} = 0.6488$) and is statistically indistinguishable from zero in the remaining three. A single-regime GEPU coefficient averages an active effect against three null ones, producing a miscalibrated hedge ratio worse than the baseline; structural breaks alone cannot exploit the directional GEPU signal within each window. Only the joint specification resolves both problems, allowing regime-specific GEPU coefficients to reflect true within-regime transmission. This finding is consistent with Shahzad et al. (2017), who document that GEPU's influence on commodity markets is nonlinear and regime-concentrated — a pattern that by construction requires both extensions to be applied jointly to deliver hedging value.

Collectively, the results confirm that the incorporation of GEPU can improve Bitcoin hedging effectiveness across all three asset classes, but its effect is modest and conditional — an additional 0.42–0.89 percentage points of variance reduction relative to the benchmark, depending on the asset pair, and negative when applied without structural break accommodation for bonds and commodities. Importantly, these gains survive realistic trading frictions: as shown in Appendix B, net hedging effectiveness remains positive across all three pairs even at a one-way rebalancing cost of 50 basis points, indicating that the documented benefits are economically — not just statistically — meaningful. That net hedging effectiveness barely erodes from the no-cost benchmark is itself informative: the incremental gain originates in the slow-moving long-run correlation component, so GEPU conditioning and structural-break accommodation reset the level of the optimal hedge ratio across regimes rather than adding high-frequency churn to it. Because daily rebalancing cost is driven almost entirely by the short-run component common to all four specifications, the extra turnover attributable to the full model is marginal — which is precisely why its variance-reduction advantage survives intact even at a punitive 50 bps one-way cost.

Ultimately, however, the dominance of speculative sentiment and network effects in Bitcoin's price dynamics, relative to macroeconomic fundamentals, ultimately limits the incremental value of any macro-conditioning framework. Nevertheless, the consistent superiority of the full DCC-MIDAS-X with structural breaks across all three pairs provides clear support for H4 and establishes that regime-aware GEPU conditioning represents the most reliable approach to Bitcoin hedging currently available in this study’s empirical setting, consistent with the modelling logic of Fang et al. (2019) and Xiong et al. (2024).

Finally, to confirm that these improvements are statistically meaningful rather than an artefact of in-sample fitting, the author applies the Diebold–Mariano and Clark–West tests to the out-of-sample loss differentials between the full model and the baseline. The results are reported in Table 4.12.

Table 4.12: Statistical significance of the hedging improvement

Asset pair	DM statistic	DM p-value	CW statistic	CW p-value	Significant (CW)
BTC–MSCI World	2.8721	0.0019***	2.1962	0.0140**	Yes (5%)
BTC–S&P GSCI	2.0173	0.0154**	2.7388	0.0031***	Yes (1%)
BTC–PIMCO	1.9338	0.1752	−1.4166	0.6615	No

Notes: DM = Diebold–Mariano (1995); CW = Clark–West (2007) adjusted MSPE, preferred for nested model comparison. The null is equal predictive accuracy; rejection favours the full model. The superscripts ***, ** and * denote statistical significance at 1%, 5% and 10% levels, respectively.

(Sources: Author’s analysis)

Table 4.12 provides the statistical validation for the hedging gains reported in Table 4.11. For BTC–MSCI and BTC–S&P GSCI, both the Diebold–Mariano and Clark–West tests reject the null of equal predictive accuracy in favour of the full DCC-MIDAS-X model with structural breaks. This confirms that the higher hedging effectiveness for the equity and commodity pairs is not merely a mechanical increase in variance reduction, but reflects statistically meaningful improvements in hedge-ratio forecasting. The evidence is particularly strong for BTC–S&P GSCI, where the Clark–West test is significant at the 1% level, indicating that regime-aware GEPU conditioning provides a clear forecasting advantage when Bitcoin's co-movement with commodities is structurally unstable.

By contrast, the BTC–PIMCO pair does not show statistically significant improvement. This does not necessarily imply that the framework fails for bonds; rather, it reflects the weak underlying Bitcoin–bond relationship. As shown in Table 4.2, the unconditional Bitcoin–PIMCO correlation is only 0.101, leaving limited systematic co-movement for any dynamic hedge ratio to exploit. In this case, GEPU conditioning and structural breaks may still improve model fit, but the resulting hedging gain is too small to be statistically distinguishable. Overall, Table 4.12 refines the conclusion of H4: the full regime-aware DCC-MIDAS-X model delivers statistically reliable hedging improvements for equity and commodity exposures, while its benefit for bonds is economically limited by the already weak Bitcoin–bond dependence structure. This result strengthens the main conclusion of the thesis that Bitcoin's hedging value is asset-class specific and regime-dependent, rather than universal.

Summary of Chapter 4

Chapter 4 evaluates whether GEPU explains Bitcoin's long-run volatility, dynamic correlations, and hedging performance. The Bai–Perron test identifies three GEPU break dates — May 2019, January 2021, and July 2024 — dividing the sample into four regimes: the trade war, pandemic, monetary tightening cycle, and post-ETF institutional era.

The GARCH-MIDAS-X results show that GEPU affects long-run volatility differently across asset classes. Its effect is negative for Bitcoin θ_X =-5.38, suggesting that rising uncertainty reduces persistent Bitcoin volatility as investors cut speculative exposure, while it is positive for equities, commodities, and bonds, consistent with the standard risk-premium channel. The break coefficients further show that this transmission is regime-dependent: Bitcoin's negative GEPU effect reversed during the pandemic as speculative interest surged, before stabilizing in the post-ETF period.

The DCC-MIDAS-X results show a similar asymmetry in correlations. Rising GEPU weakens Bitcoin's co-movement with equities θ_X =-1.80, but strengthens its correlations with commodities (0.71) and bonds (0.47). This implies that Bitcoin's hedging value improves most clearly for equities during high-uncertainty periods, while its role against commodities and bonds weakens under the same conditions. The break coefficients confirm that these relationships are structurally unstable, especially during the trade war, pandemic, and post-policy-normalization regimes.

The full DCC-MIDAS-X model with structural breaks achieves the highest variance reduction across all three pairs — 17.05% for BTC–MSCI, 3.03% for BTC–PIMCO, and 2.18% for BTC–GSCI. Regime awareness drives most of the equity improvement, while for bonds and commodities the hedging gain emerges mainly when GEPU conditioning and structural breaks are modeled jointly. These results support H1–H4 and clarify the central contribution of this thesis: it extends the GEPU-conditioned Bitcoin hedging framework of Fang et al. (2019) by incorporating structural breaks, and extends the structural-break DCC-MIDAS-X logic of Xiong et al. (2024) to cryptocurrency hedging. More importantly, it decomposes the source of hedging improvement into GEPU effects, structural-break effects, and their joint interaction, showing that Bitcoin's hedging value is asset-class specific and regime-dependent rather than universal.

CHAPTER 5: CONCLUSIONS AND RECOMMENDATIONS

5.1 Conclusions of findings

5.1.1 Summary of key findings

Global economic policy uncertainty carries significant information about the long-run volatility of Bitcoin, equities, commodities, and investment-grade bonds, although its effect differs sharply across asset classes. In the regime-aware GARCH-MIDAS-X volatility equation, the GEPU coefficient for Bitcoin is negative and significant ($\theta^{V}_{GEPU} = -5.3840$), indicating that higher policy uncertainty is associated with lower persistent Bitcoin volatility over the sample period. This pattern is in line with He et al. (2024), who show that economic policy uncertainty adversely affects Bitcoin's long-term value, suggesting that high-uncertainty periods may induce de-risking rather than speculative inflows. However, this mechanism cannot be directly confirmed without investor-flow or trading-volume data. More importantly, this result contrasts with Fang et al. (2019), who found a positive GEPU–Bitcoin volatility relationship in their pre-2018 sample, suggesting that Bitcoin's uncertainty transmission channel may have changed as the asset shifted from a retail-driven speculative instrument toward a more institutionally managed asset class. By contrast, MSCI World, S&P GSCI, and PIMCO all report positive and significant GEPU effects in the volatility equation, consistent with the standard Pástor and Veronesi (2013) discount-rate channel, where rising policy uncertainty increases required risk premia and feeds into higher persistent volatility. Accordingly, the sharp difference between Bitcoin (-5.38) and PIMCO (+6.83) confirms that the same macroeconomic shock can transmit through very different volatility channels depending on the asset class. These results support H1 and show that GEPU adds genuine explanatory power for long-run volatility beyond realized variance alone.

Moreover, the GEPU transmission to long-run correlations displays an equally important pattern of asset-class heterogeneity. In the DCC-MIDAS-X correlation equation, the GEPU coefficient for Bitcoin–MSCI is negative and highly significant ($\theta^{C}_{GEPU} = -1.7977$), indicating that Bitcoin's long-run co-movement with global equities declines when policy uncertainty rises. In practical terms, Bitcoin becomes more useful as an equity hedge during high-uncertainty episodes, when institutional investors have a stronger need for portfolio risk reduction. By contrast, the GEPU coefficients for Bitcoin–GSCI and Bitcoin–PIMCO are positive and significant ($\theta^{C}_{GEPU} = 0.7068$ and 0.4709), meaning that rising uncertainty pushes Bitcoin to move more closely with commodities and bonds. During these periods, commodities are exposed to demand uncertainty, bonds are exposed to monetary-policy ambiguity, and Bitcoin's diversification value against these assets weakens rather than strengthens. This sign asymmetry supports H2 and shows that GEPU affects Bitcoin's hedging role differently across asset pairs, consistent with flight-to-quality dynamics in which uncertainty reshapes capital flows and cross-asset dependence.

However, none of these relationships are stable across the full sample period. Allowing the MIDAS slope coefficients to shift across the three Bai–Perron break dates — May 2019, January 2021, and July 2024 — produces uniformly lower AIC and BIC values for every asset and every Bitcoin–asset pair, confirming that the GEPU transmission mechanism is structurally unstable across uncertainty regimes. The regime shifts are not merely statistical artefacts; they correspond to well-identified macroeconomic events and carry clear economic interpretations. For Bitcoin's long-run volatility, the suppressive effect of GEPU reversed sharply during the pandemic regime ($\theta_{X,\text{break2}} = 2.8551$, $p < 0.01$), when the extraordinary surge in both retail and institutional speculative interest meant that rising uncertainty amplified rather than compressed Bitcoin's persistent volatility. By the post-ETF regime, this reversal disappeared ($\theta_{X,\text{break3}} \approx 0$), suggesting that institutional investors had begun pricing policy uncertainty into Bitcoin more systematically. For the correlation structure, the break coefficients tell an equally coherent story: the negative baseline GEPU effect on Bitcoin–equity correlation was temporarily overridden during the trade war and pandemic regimes, when risk-off liquidation drove Bitcoin and equities to fall together. For Bitcoin–PIMCO, the flight-to-quality mechanism operated most powerfully during the trade war regime ($\theta_{X,\text{break1}} = -3.3976$) before reversing during the pandemic as the extraordinary monetary policy response simultaneously affected both asset classes. These results provide strong support for

H3 and confirm the warning of Xiong et al. (2024) that ignoring structural breaks generates systematically biased MIDAS estimates.

The bottom-line test of whether these modelling gains translate into portfolio benefits comes from the hedging effectiveness comparison. The full DCC-MIDAS-X model with structural breaks delivers the highest variance reduction across all three Bitcoin–asset pairs: 17.05% for BTC–MSCI, 3.03% for BTC–PIMCO, and 2.18% for BTC–GSCI. Decomposing the incremental improvement reveals an important nuance. For the equity pair, structural-break accommodation accounts for most of the gain, while GEPU conditioning adds only marginal improvement. For the bond and commodity pairs, the benefit of the full model comes mainly from allowing the Bitcoin–asset relationship to change across regimes; without structural breaks, the model forces different uncertainty periods into one average effect and weakens the estimated hedging signal.

Crucially, the results show that GEPU conditioning and structural-break accommodation work best when modeled jointly rather than separately. A regime-aware framework improves hedge-ratio calibration because it allows the long-run covariance structure to adjust when macroeconomic uncertainty changes. However, the absolute gains remain modest, reflecting the constraint imposed by Bitcoin's high volatility on achievable portfolio risk reduction, as emphasized by Smales (2019) and Conlon and McGee (2020). Overall, the full specification provides qualified support for H4: regime-aware GEPU conditioning improves Bitcoin hedging performance relative to the benchmark models, especially for equity and commodity exposures, while the benefit for bonds is more limited.

Collectively, these four sets of findings tell a coherent and internally consistent story. GEPU is a genuine driver of both long-run volatility and long-run correlations involving Bitcoin, but its influence is heterogeneous across asset classes and structurally unstable across macroeconomic regimes. Models that ignore either dimension — by excluding GEPU entirely, or by treating its effect as constant across the sample — produce hedge ratios that are systematically wrong in ways that matter for portfolio risk management. The regime-aware DCC-MIDAS-X framework resolves both problems simultaneously, delivering the most accurate hedge ratios across all asset classes and all uncertainty regimes examined.

5.1.2 Implications of the research

The results carry several practical implications for different categories of market participants.

For institutional portfolio managers allocating across equities, commodities, and bonds, the central takeaway is that Bitcoin's hedging value is not a single number but a function of two things: which asset class is being hedged, and which uncertainty regime the economy is currently in. Managers holding global equity portfolios should therefore treat Bitcoin as a state-dependent hedge — increasing the hedge position when uncertainty is elevated and reducing it when conditions are calm — rather than holding a fixed allocation. For commodity exposure, the benefit is smaller but still positive, so Bitcoin can play a supporting role. For investment-grade bonds, however, the evidence does not support using Bitcoin as a hedge, and managers should not expect meaningful variance reduction from doing so. This asymmetry means that a one-size-fits-all Bitcoin hedge ratio applied uniformly across a multi-asset portfolio will be miscalibrated for at least some of the underlying positions at any given time.

Moreover, the dominance of the BTC–MSCI pair in the hedging effectiveness results — delivering 17.05% variance reduction compared to 3.03% and 2.18% for the bond and commodity pairs — suggests that the primary practical application of the regime-aware framework lies in equity hedging. This finding carries a natural extension to country-level equity exposures. The MSCI World Index is a capitalization-weighted aggregate of 23 developed market indices, meaning that the estimated BTC–MSCI correlation reflects a diversified average across national equity markets with heterogeneous GEPU sensitivities. Countries whose national EPU contributes disproportionately to the global index — most notably the United States, China, Japan, and the major European Union economies, which collectively account for over 60% of the GEPU weighting (Davis, 2016) — are likely to exhibit stronger and more volatile Bitcoin co-movement than the diversified global benchmark, because the aggregation inherent in MSCI World mechanically smooths away country-specific correlation spikes that matter most for domestic portfolio managers. Consequently, institutional investors managing concentrated equity exposures in high-EPU-weight economies may find that the hedging gains documented here for the global index represent a lower bound on the achievable variance reduction at the country level — particularly during regimes where the national policy cycle diverges from the global average, generating country-specific correlation dynamics that the MSCI World aggregate cannot capture. For portfolio managers at domestic pension funds or sovereign wealth funds whose equity benchmarks are national rather than global, recalibrating the DCC-MIDAS-X framework with country-specific indices and national EPU measures therefore represents the most direct path to operationalizing the findings of this study.

For risk managers responsible for monitoring and rebalancing hedging positions, the dominance of structural break accommodation in the hedging effectiveness decomposition carries a specific operational implication. The level of GEPU on any given month matters less than whether the economy has crossed a regime boundary. The three break dates identified in this study — May 2019, January 2021, and July 2024 — all correspond to observable macroeconomic events: the collapse of US–China trade negotiations, the transition out of the acute pandemic phase, and the post-ETF institutional reconfiguration of Bitcoin's investor base. In practice, risk managers should treat GEPU less as a monthly input to be updated mechanically and more as a signal of regime change. Once the data indicate that the economy has moved into a new uncertainty regime, hedge ratios should be recalibrated using the new regime-specific parameters rather than simply adjusted according to the latest GEPU value.

For academic researchers studying Bitcoin’s evolving role in financial markets, the sign reversal of θ_X for Bitcoin’s long-run volatility is one of the most important findings of this thesis. While Fang et al. (2019) documented a positive GEPU–Bitcoin volatility relationship in the pre-2018 period, the negative coefficient in the present study suggests that Bitcoin’s response to policy uncertainty has changed as the market becomes more institutionally managed. This shift implies that Bitcoin can no longer be modelled as if its uncertainty transmission mechanism were stable over time. Instead, its volatility and correlation dynamics should be examined through a regime-aware mixed-frequency framework, such as the structural-break GARCH-MIDAS-X and DCC-MIDAS-X models used in this thesis. By allowing macroeconomic uncertainty effects to vary across structural regimes, this approach provides a more suitable framework for assets whose investor composition, regulatory environment, and market microstructure evolve rapidly.

Finally, the findings also have implications for policymakers and financial regulators. The results show that Bitcoin is no longer fully isolated from conventional financial markets, as its long-run volatility and correlations respond significantly to global economic policy uncertainty. This means that major policy shocks, such as trade tensions, fiscal uncertainty, or monetary tightening, can affect not only traditional assets but also Bitcoin's risk dynamics and its co-movement with equities, commodities, and bonds. Regulators should therefore monitor Bitcoin not only as a standalone crypto asset, but also as part of the broader financial system, especially as spot Bitcoin ETFs increase its accessibility to institutional investors. The regime-dependent results further suggest that Bitcoin's diversification benefits may weaken during periods of systemic stress, when correlation structures change most sharply. For this reason, financial stability assessments should incorporate crypto–traditional market linkages, ETF-related exposure channels, and the possibility that Bitcoin's risk-reduction role differs across uncertainty regimes.

5.2 Limitations and recommendations for further research

5.2.1 Limitations of the research

Several limitations should be acknowledged when interpreting the results of this research.

First, the sample period remains relatively short for macro-financial analysis. Although the dataset covers nine years and four uncertainty regimes, the GEPU variable is observed monthly, providing only 110 observations for the MIDAS estimation. This limits the degrees of freedom available for estimating regime-specific slope coefficients, especially in the final post-ETF regime from July 2024 to

December 2025, which contains only 18 monthly observations. As a result, findings related to this most recent regime, such as the post-ETF changes in Bitcoin's correlations with equities and bonds, should be interpreted with caution. As more post-ETF data become available, future studies can re-estimate these relationships with stronger statistical reliability.

Second, the estimation and inference design carries two related constraints. The model uses GEPU as its only low-frequency macroeconomic conditioning variable. While this single-variable design isolates the policy uncertainty channel cleanly, it means that other potentially important drivers — such as global liquidity conditions, monetary policy stance, credit spreads, or geopolitical risk indices — are not controlled for, so the estimated θ_X coefficients may partly reflect omitted variables correlated with GEPU rather than the pure effect of policy uncertainty. In addition, the two-step procedure — in which structural breaks are first identified from the GEPU series and then imposed as fixed dummies in the second stage — does not propagate break-point estimation uncertainty into the second-stage inference. This is standard practice in the DCC-MIDAS literature with structural breaks (Fang et al., 2019; Xiong et al., 2024), but it may understate the true standard errors of the MIDAS slope coefficients by 5–15% in finite samples of comparable size (Elliott and Müller, 2007). The ±3 month break-date perturbation check partially addresses this by confirming that the sign and significance of the key coefficients are robust to alternative regime datings, although a fully joint estimation of break locations and model parameters would provide more rigorous inference.

Third, the hedging effectiveness analysis is conducted at the bilateral level — each Bitcoin–asset pair is evaluated independently — rather than within an integrated multi-asset portfolio optimisation framework. In reality, institutional investors do not hedge equities, commodities, and bonds separately; they manage a combined portfolio in which the optimal Bitcoin allocation depends on the full conditional covariance matrix across all assets simultaneously. The bilateral approach, while standard in the hedging literature and consistent with the research design of Fang et al. (2019) and Bouri et al. (2017), does not capture potential interaction effects among the three hedged positions and may therefore overstate or understate the aggregate hedging benefit of Bitcoin in a diversified portfolio context.

Fourth, the study does not account for several practical frictions that would affect real-world implementation. While the out-of-sample rolling-window evaluation and the transaction-cost adjustment at 10, 25, and 50 basis points provide important robustness checks, they do not capture margin requirements for maintaining Bitcoin positions, liquidity constraints during extreme market stress when bid-ask spreads widen substantially, or the potential market impact of large institutional trades. Given that Bitcoin's market microstructure remains less mature than that of conventional asset classes — despite the significant improvements following ETF approval — these frictions could meaningfully reduce the implementable hedging gains relative to the theoretical improvements documented in the empirical results.

5.2.2 Recommendations for further research

Several directions for future research would address the limitations identified above and extend the contributions of this study.

First, incorporating multiple macroeconomic conditioning variables into the MIDAS long-run component — such as the VIX, global financial conditions indices, central bank policy rate expectations, and the Caldara and Iacoviello (2022) geopolitical risk index — would allow researchers to decompose the relative contribution of different macro drivers to Bitcoin's long-run volatility and correlations. Following the multi-covariate GARCH-MIDAS-X specification of Conrad and Loch (2015), this extension would test whether GEPU retains its significance after controlling for correlated uncertainty measures and would identify which dimensions of macroeconomic uncertainty matter most for Bitcoin's hedging properties.

Second, future research should adopt a more advanced modelling approach that estimates structural break locations and MIDAS parameters jointly within a single framework, rather than relying on the current two-step procedure in which the breaks are fixed in advance. Several advanced methods could achieve this integration.

Penalised maximum-likelihood estimation with a fused-LASSO penalty on regime-specific slope coefficients would allow breaks to be detected and parameters to be estimated in one step. Reversible-jump MCMC offers a fully Bayesian alternative by treating the number and location of breaks as unknown parameters with their own priors. Markov-switching MIDAS specifications go further by modelling regimes as latent states governed by a transition matrix, allowing the model to learn when the uncertainty environment shifts rather than imposing break dates externally. In addition, a hybrid bootstrap-copula Monte Carlo framework could be used to simulate regime-specific return and dependence structures while preserving both marginal uncertainty and cross-asset tail dependence. Such a framework would be particularly useful for stress-testing Bitcoin hedge ratios under alternative uncertainty regimes and for assessing whether the estimated hedging gains remain robust under simulated extreme-market scenarios. Together, these approaches would propagate break-point, parameter, and dependence uncertainty more directly into the inference on MIDAS slope coefficients, producing more conservative but more reliable confidence intervals for the regime-specific transmission parameters and placing the conclusions of this study on firmer statistical ground.

Third, extending the analysis from bilateral hedging to a full multi-asset portfolio optimization framework represents a natural and practically important next step. Under this approach, Bitcoin's allocation would be jointly optimized alongside equities, commodities, and bonds using the complete conditional covariance matrix estimated by the DCC-MIDAS-X model, rather than being determined independently for each pair. Mean-variance, minimum-variance, and risk-parity portfolio construction methodologies all represent feasible extensions that would provide more realistic and actionable guidance for institutional portfolio managers.

Fourth, given that the BTC–MSCI pair produces the highest hedging effectiveness across all specifications, a natural and practically significant extension would be to disaggregate the global equity benchmark into country-level indices and test whether the regime-aware framework delivers stronger hedging gains at the national level. Replacing the MSCI World Index with major country indices — such as the S&P 500 for the United States, the CSI 300 for China, the Nikkei 225 for Japan, and the EURO STOXX 50 for the Eurozone — and substituting the corresponding national EPU index of Baker et al. (2016) for the GDP-weighted GEPU measure would allow researchers to identify which national equity markets benefit most from Bitcoin hedging under policy uncertainty, and whether the country-specific EPU signal produces sharper regime identification and higher variance reduction than the aggregated global index. This decomposition is particularly relevant because the Bai–Perron break dates identified in the present study — the trade war escalation, the pandemic shock, and the post-ETF institutional shift — affected national equity markets with markedly different timing and severity, suggesting that country-specific structural breaks may yield more precise regime dating and, by extension, more accurate hedge ratios than the global breaks imposed uniformly across all pairs in this study.

Finally, exploring alternative frequency combinations — such as weekly returns with quarterly GEPU or intraday returns with daily uncertainty measures — would test whether the mixed-frequency patterns documented at the daily-monthly level generalise across different investment horizons. This is particularly relevant for institutional investors whose rebalancing frequencies range from daily for active trading desks to quarterly or annual for strategic asset allocation committees, each of whom faces a different optimal hedge ratio horizon.

Summary of Chapter 5

Chapter 5 concludes that Bitcoin's hedging value under GEPU is asset-class specific and regime-dependent. GEPU significantly affects both long-run volatility and dynamic correlations, but its impact differs across markets. Rising uncertainty reduces Bitcoin's persistent volatility while increasing volatility in conventional assets. At the same time, higher GEPU weakens Bitcoin's co-movement with equities but strengthens its links with commodities and bonds. Therefore, Bitcoin appears more effective for equity hedging during uncertain periods, while its protective value for commodities and fixed income is more limited.

A further key finding is that these relationships are structurally unstable over time. Because Bitcoin's volatility and correlations change across regimes, a model based on one fixed full-sample relationship can produce inaccurate hedge ratios. In contrast, the full DCC-MIDAS-X model with structural breaks delivers the strongest portfolio risk reduction, especially for the Bitcoin–equity pair, with a 17.05% variance reduction compared with smaller gains for bonds and commodities.

The chapter then explains what these results mean for different groups. For portfolio and risk managers, the main lesson is that there is no single Bitcoin hedge ratio that works for every situation; instead, the appropriate ratio depends on which asset is being hedged and which uncertainty regime the economy is currently in. Practitioners are therefore advised to monitor GEPU for signs of a regime change rather than focusing on its monthly level. For academic researchers, the most important result is the reversal in the direction of Bitcoin's response to uncertainty compared with earlier studies, which shows that Bitcoin has matured into an institutionally driven asset and can no longer be modelled using parameters from an earlier era.

Finally, the chapter acknowledges several limitations, including the short post-ETF sample, the single GEPU conditioning variable, the bilateral hedging design, simplified transaction costs, and the two-step treatment of structural breaks. Future research should extend the framework empirically by adding broader macro-financial variables, national EPU indices, country-level equity exposures, and full multi-asset portfolios. Methodologically, future studies could adopt more advanced quantitative models, such as fused-LASSO penalised likelihood, reversible-jump MCMC, Markov-switching MIDAS, or hybrid bootstrap-copula Monte Carlo simulations, to jointly estimate break locations and MIDAS parameters while capturing parameter uncertainty and tail dependence. Overall, Chapter 5 concludes that Bitcoin's hedging value is asset-class specific and regime-dependent, making regime-aware modelling essential for future Bitcoin risk-management research.

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APPENDIX

A. Test for structural breaks

To accommodate regime instability in the macroeconomic environment, this thesis applies the Bai–Perron (1998, 2003) multiple structural break test to the GEPU series, endogenously detecting m unknown breakpoints that partition the sample into distinct uncertainty regimes.

The optimal number of breakpoints is estimated by selecting break dates that minimize the total sum of squared residuals (SSR) across all admissible partitions:

$$\{\widehat{T_1}, \dots, \widehat{T_m}\} = \operatorname{argmin}_{T_1,\dots,T_m} \sum_{j=1}^{m+1} S(T_{j-1} + 1, T_j) \qquad (A.1)$$

where $S(T_{j-1} + 1, T_j)$ is the SSR for segment j, with $T_0 = 0$ and $T_{m+1} = T$. Global minimisers are obtained via the dynamic-programming algorithm of Bai and Perron (2003). The number of breaks is determined sequentially using the sup-*F* statistic for ℓ + 1 breaks against ℓ breaks:

$$F(\ell + 1|\ell) = \frac{[S(\widehat{T_1}, \dots, \widehat{T_\ell}) - S(\widehat{T_1}, \dots, \widehat{T_{\ell+1}})]}{S(\widehat{T_1}, \dots, \widehat{T_{\ell+1}})} \cdot \frac{T - (\ell + 1)k}{k} \qquad (A.2)$$

where k is the number of parameters allowed to shift and T the total number of observations. The null of ℓ breaks is rejected in favour of ℓ + 1 when the statistic exceeds the Bai–Perron (1998) critical value, yielding the estimated number of breaks $\widehat{m}$. Following Bai and Perron (2003), the minimum segment length is set to 15% of the sample to ensure reliable estimation, and break dates are reported with 95% confidence intervals.

B. Hedging effectiveness of Bitcoin after transaction-cost adjustment

Net HE adjusts the no-cost index for daily rebalancing: c = one-way cost (basis points) applied to changes in the optimal hedge ratio.

Model	**HE (no cost)**	**c = 10 bp**	**c = 25 bp**	**c = 50 bp**
	Panel A: BTC–MSCI World			
DCC-RC (no SB)	0.161593	0.161589	0.161583	0.161572
DCC-RC (with SB)	0.169847	0.169838	0.169825	0.169803
DCC-X (no SB)	0.162846	0.162841	0.162834	0.162822
DCC-X (with SB) ★	**0.170464**	**0.170461**	**0.170455**	**0.170446**
	Panel B: BTC–S&P GSCI			
DCC-RC (no SB)	0.017616	0.017614	0.017611	0.017607
DCC-RC (with SB)	0.018366	0.018338	0.018302	0.018291
DCC-X (no SB)	0.01468	0.014679	0.014679	0.014679
DCC-X (with SB) ★	**0.02184**	**0.021836**	**0.021835**	**0.021832**
	Panel C: BTC–PIMCO Corp Bond			
DCC-RC (no SB)	0.022042	0.022029	0.02201	0.021978
DCC-RC (with SB)	0.030112	0.030102	0.030086	0.030059
DCC-X (no SB)	0.019656	0.019643	0.019625	0.019595
DCC-X (with SB) ★	**0.030251**	**0.030241**	**0.030226**	**0.030201**

Notes: Net HE accounts for daily rebalancing costs; the no-cost column is carried over from Table 4.11. ★ = best specification. DCC-RC = DCC-MIDAS with realized correlations; DCC-X = DCC-MIDAS-X with GEPU; SB = Bai–Perron (2003) structural breaks.

(Sources: Author's analysis)

C. Model specification and forecast comparison tests

Panel A: Likelihood-ratio tests for model extensions.

Panel B: Diebold–Mariano and Clark–West forecast comparison tests.

Panel A: Likelihood-Ratio Tests (H_0: restricted model adequate; χ^2(df))			
Test description	**LR Statistic**	**df**	**p-value**
BTC: GARCH-MIDAS-X vs RV (GEPU adds power)	8.4017	2	0.014983**
MSCI: GARCH-MIDAS-X vs RV	23.1415	2	0.000009***
GSCI: GARCH-MIDAS-X vs RV	17.8503	2	0.000133***
PIMCO: GARCH-MIDAS-X vs RV	7.5226	2	0.023253**
BTC: X+SB vs X (regime shifts improve fit)	19.3421	6	0.003623***
MSCI: X+SB vs X	88.5219	6	0.000000***
GSCI: X+SB vs X	72.1296	6	0.000000***
PIMCO: X+SB vs X	47.0484	6	0.000000***

Panel B: Forecast Comparison — full model (DCC-X+SB) vs baseline (DCC-RC, no SB)		
Test	**Statistic**	**p-value**
DM test — BTC–MSCI	2.8721	0.0019***
CW test — BTC–MSCI	2.1962	0.0140**
DM test — BTC–S&P GSCI	2.0173	0.0154**
CW test — BTC–S&P GSCI	2.7388	0.0031**
DM test — BTC–PIMCO	1.9338	0.1752*
CW test — BTC–PIMCO	-1.4166	0.6615

Notes: LR statistic = $2(\ell_{unrestricted} - \ell_{restricted}) \sim \chi^2(df)$. DM = Diebold–Mariano (1995); CW = Clark–West (2007) adjusted MSPE (preferred for nested models). *, **, *** denote 10%, 5%, 1% significance.

(Sources: Author's analysis)

D. Residual diagnostics - Full GARCH-MIDAS specification (RV + GEPU + SB)

Each value is the test statistic with p-value in parentheses.

Q(p) = Ljung–Box on standardized residuals; Q²(p) = Ljung–Box on squared residuals; ARCH LM(p) = Engle (1982) test. p > 0.05 confirms model adequacy.

Asset	Q(10)	Q(20)	Q²(10)	Q²(20)	ARCH LM(10)
Bitcoin	14.25(0.162)	32.53(0.038)	9.73(0.464)	15.06(0.773)	9.91(0.449)
MSCI World	26.10(0.004)	35.13(0.019)	5.98(0.817)	8.40(0.989)	6.07(0.809)
S&P GSCI	12.67(0.243)	30.79(0.058)	17.08(0.073)	21.39(0.374)	16.48(0.087)
PIMCO Corp Bond	12.40(0.259)	29.62(0.076)	17.10(0.072)	21.33(0.378)	16.23(0.093)

Notes: Diagnostics reported for the full specification (RV + GEPU + Bai–Perron structural breaks), K = 12 MIDAS lags. p > 0.05 indicates no residual serial correlation or ARCH effects.

(Sources: Author's analysis)